Supply Networks and Market Power in U.S. Poultry Procurement

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Abstract

This paper examines how supplier networks and firm ownership shape pricing and disruption losses. I reconstruct the U.S. poultry procurement network from shipment records and estimate a demand-and-supply model with joint pricing across each processor's plants. The baseline price–cost margin is 33 percent of the selling price. Wholesale buyers are 1.67 times as price sensitive as retail buyers and 3.33 times as sensitive as food-service buyers. A 20 percent raw-poultry cost increase reduces total surplus by 17.7 percent and lowers buyer and processor surplus. Merging the third- and fourth-largest processors raises prices by 6.9 percent, increasing processor surplus while reducing buyer and total surplus. Regional losses vary with procurement composition. Controlling for income, education, and population, a 10-percentage-point higher baseline wholesale procurement share, which is a proxy for business-serving demand, is associated with procurement losses 6 percent larger in pounds and buyer-surplus losses 4 percent larger in dollars under the input-cost shock.

JEL Codes: C45, C51, L13, L66, Q13, R41.
Keywords: procurement networks, local market structure, buyer composition, market power, supply disruptions, poultry.

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1 Introduction

How do supplier networks and firm ownership determine who bears the burden of a supply disruption? Buyers can adjust by switching suppliers, but the availability of alternative suppliers and the ownership of those alternatives jointly determine how prices, procurement, and surplus respond. In some cases, the alternative supplier may be another plant that is owned by the same firm. Measuring the burden therefore requires information about both physical sourcing relationships and the firms that control the available suppliers.

I study these mechanisms in U.S. poultry procurement, where production is geographically concentrated and organized by vertically coordinated, multi-plant firms. Under production contracts, integrators supply chicks, feed, and veterinary services to contract growers, while maintaining ownership of the birds and processing them in their own plants; 97 percent of broilers are raised on contract (Vukina, 2001; MacDonald, 2014). Winter Storm Uri illustrates the stakes: in the week ending February 20, 2021, preliminary federal data showed U.S. broiler slaughter 43.9 percent below its level a year earlier (USDA ERS, 2021). Downstream buyers' ability to replace purchases from affected plants depends on their other sourcing options and on the firms pricing those options.

To examine how this organization of supply shapes competition and shock exposure, I first reconstruct the local supplier network from shipment records, then estimate buyer substitution and processor pricing, and finally compare supply disruptions with a change in ownership. The comparison distinguishes changes in physical sourcing options and costs from changes in pricing incentives. It also asks which procurement markets bear larger losses. Demand reaches processors along two routes: retailers procure primarily for household grocery demand, which I call household-serving procurement, while wholesalers procure primarily for food-service and institutional customers, which I call business-serving procurement.

Constructing the network requires recovering the procurement channels behind shipment records. The Commodity Flow Survey (CFS) identifies the origin establishment and destination ZIP code but does not identify the receiving establishment. I construct the first shipment-based representation of the U.S. poultry procurement network from approximately 73,000 CFS records for 2007, 2012, and 2017, linked to Census establishment, ownership, production, and trade data. I use destination business composition to label unambiguous processor-to-retailer (PR) and processor-to-wholesaler (PW) shipments, then train a random forest on the labeled shipments to classify shipments to mixed destinations; this method reaches about 96 percent accuracy on a held-out validation sample. PR shipments primarily carry household-serving procurement, and PW shipments primarily carry business-serving procurement. A third shipment label, *From W*, marks shipments originating at wholesalers

and describes local distribution from wholesale onward. Establishment linkages add plant costs and firm ownership. Together, the records identify local supplier choice sets, procurement shares, and the firms pricing the available plant-products.

The constructed network reveals a sparse set of sourcing relationships: annual plant-to-county network densities average 10 percent for retailer-bound shipments and 14 percent for wholesaler-bound shipments. Processor-to-wholesaler shipments are about 3.5 times as large as processor-to-retailer shipments and travel 48 percent farther. A plant's wholesaler-bound shipments are also concentrated on fewer destinations; the destination Herfindahl–Hirschman index (HHI), based on each plant's shipment shares across destinations, averages 0.64 for PW shipments and 0.24 for PR shipments (Table 1). These patterns describe large purchases assembled at wholesale nodes, followed by local distribution. I measure each county's business-demand orientation using its baseline wholesale procurement share, or PW share: the quantity-weighted share of processor-origin poultry procurement shipped to wholesalers. This measure summarizes the delivery channel and serves as a proxy for the demand organized through it.

To measure competition on this network, I estimate a regional demand-and-supply model following Berry (1994) and Berry, Levinsohn, and Pakes (1995). A product is a processing plant's fresh or frozen poultry, and a market is a destination county in a survey year. Plants in the sample specialize in one commodity category (broiler, processed poultry, or turkey and other poultry), which defines the nests. Random-coefficients nested logit demand allows substitution to differ across products and across three types of procurement decision makers: retail, wholesale, and food-service buyers. Buyer types describe who chooses the supplier. A food-service operator may choose its processor while a wholesaler handles delivery, so the three buyer types do not reduce to the two shipment routes. On the supply side, each processor jointly prices its products across its plants under multiproduct Bertrand competition, accounting for sales retained within its portfolio when buyers switch plants. Marginal cost depends on plant costs, scale, and shipment distance. Because prices respond to unobserved quality and matching, I instrument with plant cost and scale measures and rival-product characteristics and estimate demand and cost jointly by two-step GMM.

Using this framework, I find substantial processor pricing power over downstream buyers. The baseline price–cost margin, defined as the difference between price and marginal cost divided by price, is approximately 33 percent, and the average plant-product own-price elasticity is -3.14. Wholesale buyers are 1.67 times as price sensitive as retail buyers and 3.33 times as price sensitive as food-service buyers. Distance affects both sourcing preferences and supply costs. Limited local substitution and common plant ownership help account for the recovered margins. These estimates concern processors' sales to downstream procurement

buyers; the model measures one layer of the chain rather than the margins of retailers or wholesalers.

I then use the estimated model to compare four changes in the procurement environment. Three remove sourcing options or raise costs; the fourth changes ownership while leaving plants and costs in place. In each experiment, I recompute short-run equilibrium prices, procurement, and surplus as buyers reallocate purchases and firms adjust prices. First, I remove the plant-products of ten counties with flooding histories, reflecting the kind of outage that followed Hurricane Florence in September 2018 (Stayer, 2018). Total surplus falls 4.8 percent, even though average price rises only 0.3 percent; buyers redirect purchases to surviving plants, but the loss of sourcing choices reduces procurement value. Second, I raise effective distance by 20 percent on every route, a trading-cost shock motivated by the 2018 spike in refrigerated-truck rates (DAT Solutions, 2018; Tyson Foods, 2018). Prices rise 1.8 percent, procurement falls 4.1 percent, and total surplus falls 4.6 percent. Third, I raise each plant's raw-poultry cost by 20 percent, an input-cost shock motivated by the 2008 feed-cost surge (Pilgrim's Pride, 2008, 2009). The imposed change raises marginal cost by 12.3 percent; in the new equilibrium, prices rise 7.9 percent, procurement falls 14.6 percent, processor surplus falls 20.7 percent, and total surplus falls 17.7 percent, the largest loss of the four experiments. All changes are relative to the common model-implied baseline. The historical episodes motivate the experiments, which are stylized simulations rather than estimates of those events.

The ownership experiment merges the third- and fourth-largest firms while holding plants, products, and marginal costs fixed. It isolates the common-ownership pricing effect of acquisitions such as Pilgrim's Pride's late-2006 purchase of Gold Kist (Pilgrim's Pride, 2007a,b). Prices rise 6.9 percent, buyers lose $2.18 billion annually, and processors gain $0.90 billion. The remaining $1.28 billion is deadweight loss, accounting for 59 percent of the buyer loss. Input-cost inflation and the merger therefore raise prices by similar amounts but move processor surplus in opposite directions. Both reduce buyer and total surplus; the merger also transfers part of the buyer loss to processors.

To examine the regional burden, I compare county outcomes with different baseline PW shares, controlling for income, education, and population. These are post-estimation associations among model-implied outcomes. A 10-percentage-point higher baseline PW share is associated with 16 percent more baseline procurement and 11 percent more total surplus. Under the common 20 percent raw-poultry cost increase, a 10-percentage-point higher baseline PW share is associated with a procurement loss that is 6 percent larger in pounds and a buyer-surplus loss that is 4 percent larger in dollars. The additional price response is not statistically distinguishable from zero. More business-oriented procurement

markets are larger at baseline and lose more purchases and surplus under higher costs, making prices an incomplete measure of their burden.

The paper contributes to three areas of research. First, it measures seller-side competition in the downstream procurement layer of a major food industry. Research on food markets examines market power, vertical pricing, and productivity (Sexton, 2013; Çakır and Balagtas, 2012; Villas-Boas, 2007; Bonnet and Dubois, 2010; Weng, Vukina, and Zheng, 2015), while recent meatpacking research studies processors' buyer power in livestock procurement (Garrido et al., 2024; Moschini and Smith, 2026). Here, I used shipment links and plant ownership to identify the processors competing for each destination's purchases; these data support my estimation of suppliers' pricing power over downstream buyers. These estimates concern seller-side market power in downstream poultry procurement, whereas the meatpacking studies above concern buyer-side market power in upstream livestock procurement.

Second, the paper connects observed shipment networks to spatial competition with heterogeneous business demand. Atalay et al. (2011) and Bernard, Moxnes, and Saito (2019) study production links and their geographic organization. Ganapati (2025) studies wholesalers' sourcing, distribution networks, scale economies, and market power. I examine processor pricing on the plant-to-county poultry procurement network, combining local supplier choice sets with differences in buyer substitution and common ownership across plants. Spatial and business-to-business demand models (Miller and Osborne, 2014; Wollmann, 2018) provide the tools for estimating these responses. Linking the observed network to the pricing system yields margins that vary across plants and markets, and also shows equilibrium adjustment through both sourcing and prices.

Third, the paper compares supply disruptions and consolidation within a common estimated procurement equilibrium. While evidence on the transmission of disruptions emphasizes production links (Barrot and Sauvagnat, 2016; Boehm, Flaaen, and Pandalai-Nayar, 2019; Carvalho et al., 2021), merger simulation measures the price effects of ownership changes (Nevo, 2000b; Miller and Weinberg, 2017). The shared network and model make it possible to compare losses of sourcing options and higher costs with a merger that retains physical suppliers but changes which firm sets the prices of supplies. This comparison shows how similar price increases can accompany opposite changes in processor surplus and different losses to downstream buyers. The county comparisons further relate those losses to the orientation of procurement markets to business demand.

The rest of the paper proceeds as follows. Sections 2 and 3 describe the market setting and linked data. Section 4 presents the model and estimation. Sections 5 and 6 report estimates and counterfactuals. Section 7 concludes.

2 Institutional Background and Market Definition

2.1 The Procurement Layer of the Poultry Supply Chain

Poultry production and final consumption occupy different locations, but those endpoints do not reveal the market in which processors sell their output. The procurement layer consists of the shipment relationships connecting particular processing plants to downstream markets, the quantities purchased through each relationship, and the firms that own the supplying plants. Reconstructing this layer from shipment records is the paper's first empirical contribution. Linked shipments and firm ownership identify local supplier choice sets and pricing portfolios.

The chain begins with contract growers, who raise birds owned by the integrators and deliver them to nearby processing plants; there is no spot market for live birds. Subsection 2.4 describes the contracts. I distinguish three layers from the plants onward: production at processing plants, procurement through retailers and wholesalers, and final use by households and food-service or institutional customers. The chain thus reads growers, plants, retailers or wholesalers, and households or food service. Let P, R, W, F, and H denote processors, retailers, wholesalers, food service, and households. These are two largely parallel procurement channels with different customer bases. A flow-conserving illustration shows the distinction between procurement and final use; with processor output normalized to 100, 55 units go directly to retailers and 45 to wholesalers; wholesalers send 5 units to retailers and 40 to food service, leaving 60 units for households. The numbers illustrate the three-layer accounting rather than estimating national channel shares. Figure 1(a) draws the chain and these flows.

The descriptive data also include wholesaler-origin downstream shipments, labeled *From W*. They pool deliveries to retail and food-service destinations. Together, the processor and wholesaler records describe larger upstream wholesale purchases and smaller downstream deliveries. They do not identify a complete shipment history for the same physical batch. Moreover, the establishment receiving a delivery need not control the upstream sourcing decision. A food-service company or purchasing organization can select the processor while a wholesaler receives and distributes the product. The model therefore retains food-service, retail, and wholesale procurement decision-makers, even though processor-origin shipments are classified into PR and PW.

2.2 Geography and Local Supplier Relationships

Figure 2 compares production, procurement, and consumption across the 48 contiguous states. Production is concentrated in the South and Southeast, whereas large states such as California, Texas, Florida, and New York are prominent consumption markets due to

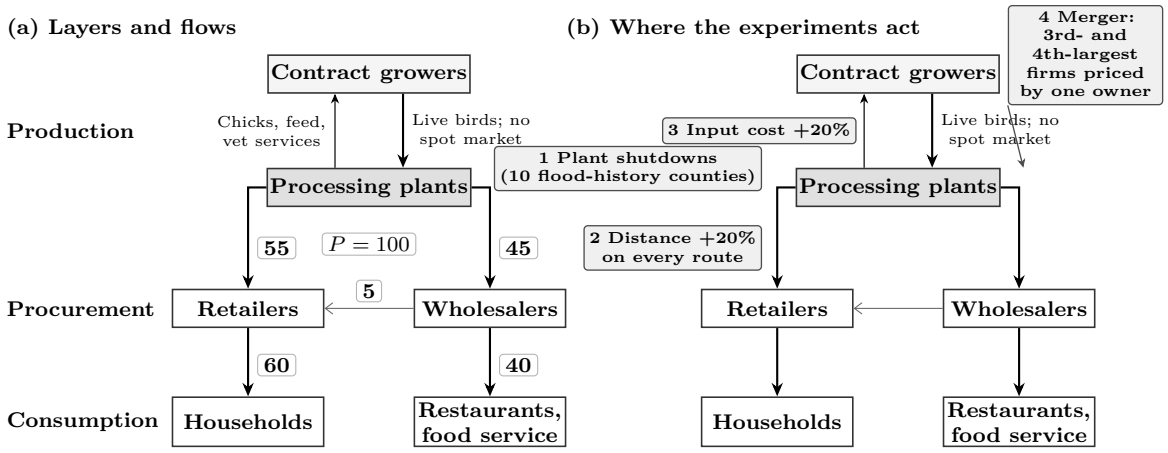


Figure 1: The Poultry Supply Chain, the Procurement Layer, and the Counterfactual Experiments
Notes: Panel (a) shows the three layers of Section 2; integrators supply chicks, feed, and veterinary services to contract growers and process the birds in their own plants, and there is no spot market for live birds. The boxed quantities are the flow-conserving illustration of Subsection 2.1 (processor output normalized to 100), not estimates of national channel shares. Panel (b) marks where the four experiments of Section 6 act: experiments 1–3 are shocks to the chain that remove sourcing options or raise costs; experiment 4 is a change in ownership that leaves plants and costs in place.

household and business demand. The producing states also take most Atlantic hurricane landfalls (Figure A9). The displayed production layer sums to 46.01 billion pounds of 2017 ready-to-cook weight. Final consumption sums to 34.71 billion pounds in retail-weight equivalents, comprising 19.34 billion pounds for households and 15.37 billion for food service. The procurement allocation is scaled to 39.11 billion pounds and retains the state shares in pooled survey-year procurement. Because these quantities refer to different stages and weight bases, I compare their normalized state distributions.

The Pearson correlation of production and consumption shares is 0.16, compared with 0.69 for production and procurement and 0.71 for procurement and consumption. Procurement is associated with both the supply base and the geography of downstream purchases. These associations describe where procurement activity is located; they are not a measure of shipping efficiency or physical distance. Figure 3 separates PR and PW procurement. The geographic processing layer marks 164 establishments in the FSIS category of at least 10 million birds per 360 days: 77 plants of the four largest firms and 87 plants of other firms. A plant marker identifies a location, whereas firm ownership determines which products are priced jointly.

State totals cannot identify the competitors serving a particular county. Figure 4 holds both plant output and county procurement fixed while changing the shipment links. Plants A and B, owned by separate firms, each produce 100 units, and counties X and Y each procure 100. In the first network, A serves only X and B serves only Y. In the second, each plant sends 50 units to each county. Closing A removes X's only observed supplier in the first network

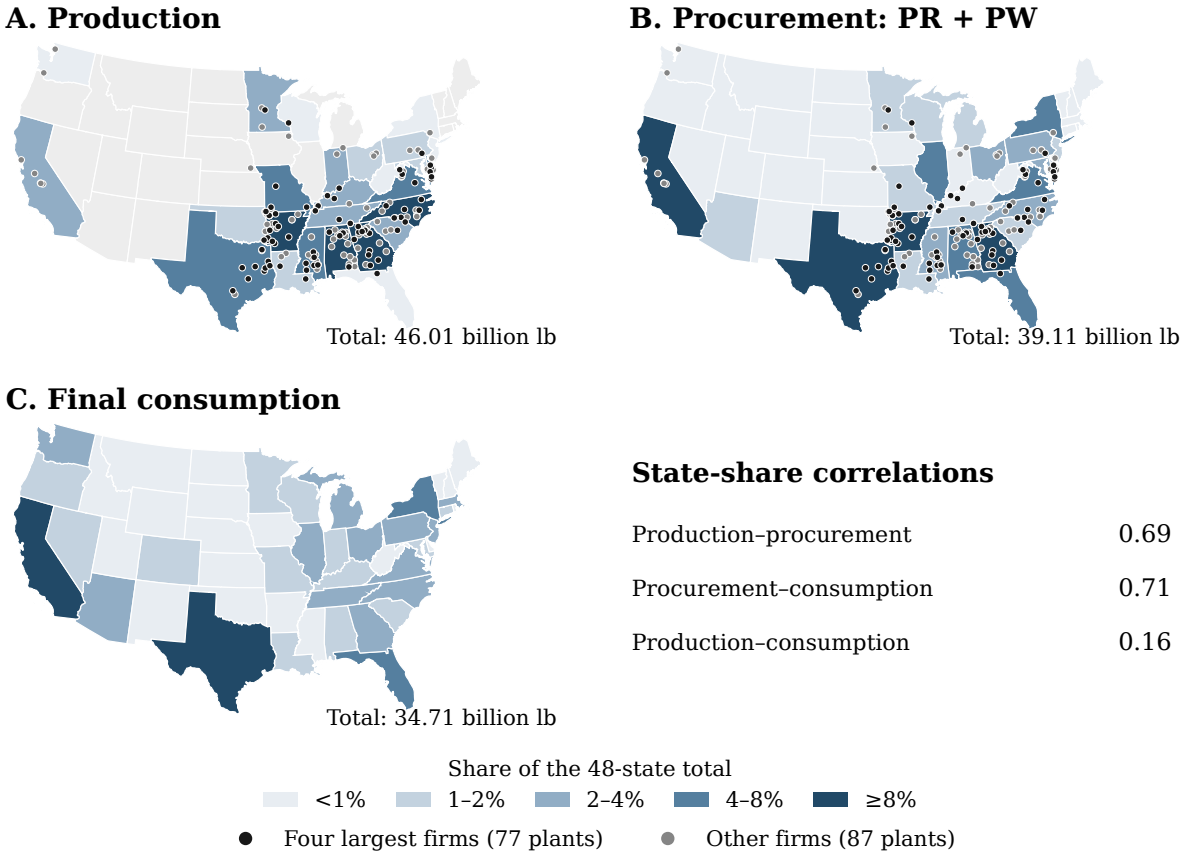


Figure 2: Production, Procurement, and Final Consumption Have Different Geographies

Notes: State shares across the 48 contiguous states, using a common color scale. Production is 2017 ready-to-cook weight; consumption is the 2017 retail-weight-equivalent allocation. Procurement preserves state shares in pooled 2007, 2012, and 2017 PR and PW shipments and scales their combined total to 39.11 billion pounds. Its geography is therefore a 2017-scaled allocation. The 22 gray production states have unreported entries, coded zero in the displayed shares and correlations. The dots identify 164 plants in the FSIS slaughter-volume category of at least 10 million birds per 360 days; 77 belong to the four largest firms and 87 to other firms. Plant coordinates are matched directory locations. Sources: the state totals and FSIS plant inputs underlying the current shipment-network data construction.

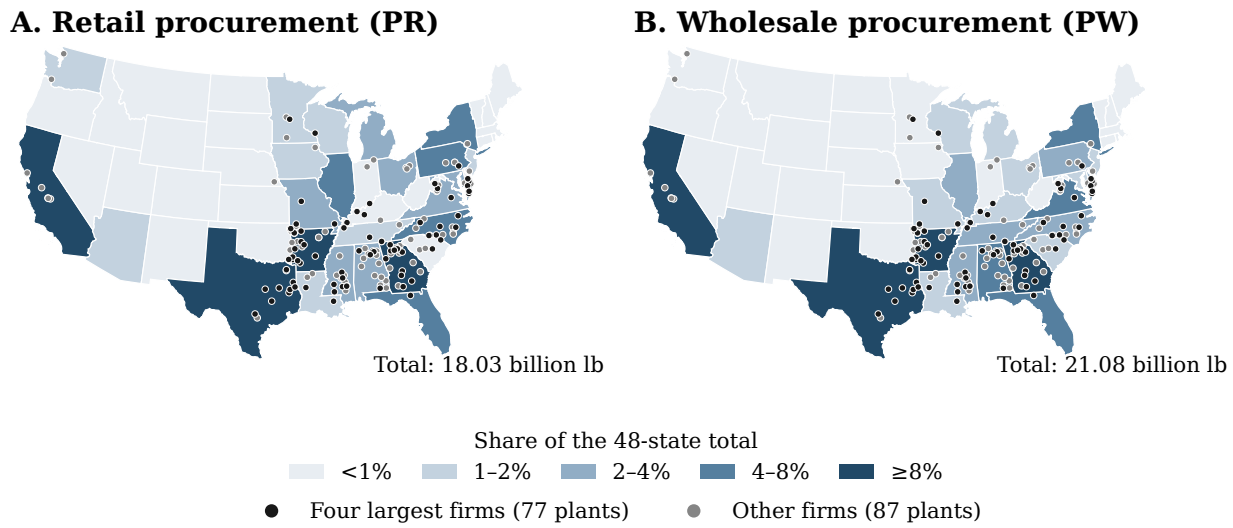


Figure 3: Retail and Wholesale Procurement Place Demand in Different Locations

Notes: State shares of processor-to-retailer (PR) and processor-to-wholesaler (PW) procurement, with a common color scale and the same processing-plant locations. Each channel's shares are normalized within that channel. The totals, 18.03 and 21.08 billion pounds, preserve pooled survey-year shares under the common 2017 scaling used in Figure 2. PR primarily represents household-serving procurement; PW brings business-serving demand to processing plants. The locations of downstream procurement and of final consumption need not coincide.

but removes half of each county's initial supply in the second. An A–B merger combines firms that serve separate counties in the first network and locally competing suppliers in the second. Identical geographic totals thus conceal different initial shock exposures and different ownership overlaps. Demand substitution and firm pricing determine the ensuing equilibrium adjustment.

2.3 Pricing Institutions in Poultry Procurement

How poultry is priced matters for how competition should be modeled. Here, spot sales refer to processed poultry sold by processors to downstream customers, whereas the earlier discussion concerns live birds raised under production contracts. Contemporaneous public materials describe a thin downstream spot layer alongside a much larger agreement-based layer. A 2016 private antitrust complaint characterizes 10–20 percent of broiler output as spot sales, 50–70 percent as sales under customer contracts, and 17–20 percent as exports (In re Broiler Chicken Antitrust Litigation, 2016, para. 84). Because this paper studies domestic transactions, a midpoint normalization within the first two categories gives a rough spot-to-nonspot volume benchmark of 15 : 60, or approximately 1 : 4. This is a range-based calibration rather than an official national time series, and "nonspot" is used here as an umbrella for recurring customer agreements, whether short- or long-term. The thinner spot

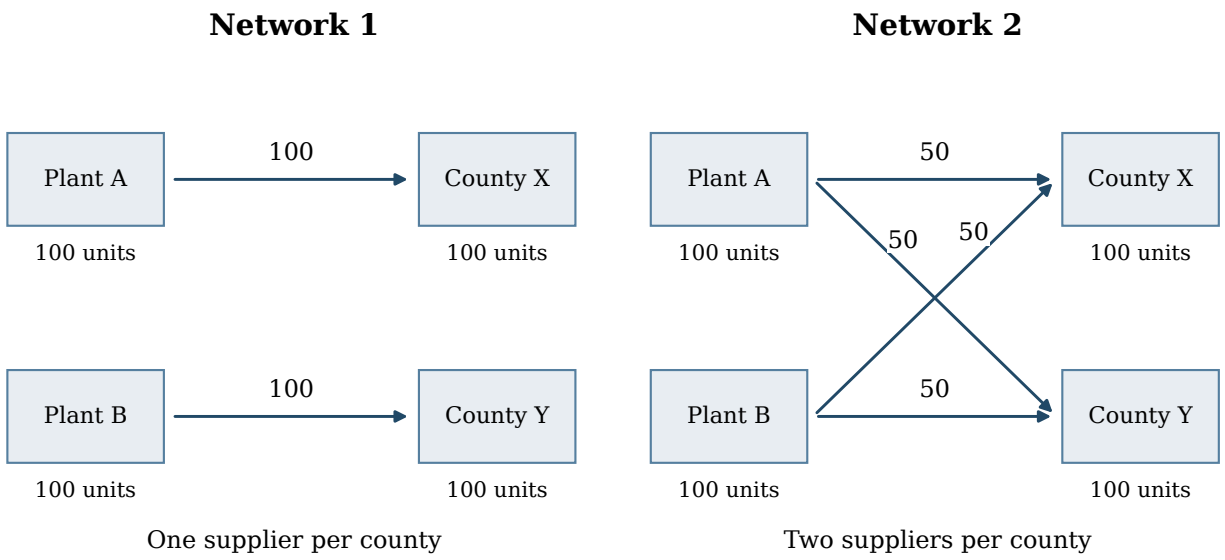


Figure 4: Identical Endpoint Totals Can Imply Different Local Competition and Shock Exposure
Notes: A thought experiment with two plants and two county markets. Both plants produce 100 units and both counties procure 100 units in either network. In Network 1, a shutdown at A removes X's only incumbent supplier, and A and B have no overlap in their served markets. In Network 2, each county initially purchases 50 units from each plant: a shutdown at A removes half of each county's incumbent supply, and a merger eliminates competition between two suppliers in both markets. Arrows show initial allocations, before any equilibrium substitution.

layer chiefly covers surplus commodity loads sold for prompt delivery; wholesalers and traders participating in that layer can redirect purchases across suppliers on short notice. Spot versus nonspot describes the pricing arrangement, while PR versus PW describes the shipment channel; wholesalers can participate in recurring agreements as well as spot purchases.

The small volume of the spot layer does not imply a small pricing role, and nonspot does not mean fixed-price. Sanderson Farms reported that prices under many long-term customer contracts fluctuated with quoted commodity prices, that realized prices commonly incorporated customer-specific discounts to those quotations, and that the choice of pricing benchmark was largely customer driven; it also reported that these agreements generally did not commit either party to a fixed quantity (Sanderson Farms, 2017). Wayne Farms similarly reported that most of its revenue arose under agreements whose formulas referenced prices reported by Urner Barry, a commercial publisher of poultry and other protein-market price quotations, with fresh-chicken prices linked to daily, weekly, or monthly quotations plus or minus a negotiated adjustment (Wayne Farms, 2015). Urner Barry is therefore a price-reporting benchmark rather than a trading exchange; quotations formed in the spot layer provide price discovery, while indexation and recurring bids or negotiations transmit that signal into a much larger nonspot segment. At the annual frequency used below, a

buyer's transaction price is not predetermined merely because the purchase occurs under a supply agreement; it can move with the benchmark and with renegotiated discounts or markups. This is a substantive contrast with beef, where mandatory livestock price reporting underpins both the public price series and the recent structural literature on packer conduct (Adjemian et al., 2016; Garrido et al., 2024). Poultry is thus a test case for measuring market power in a major meat industry without mandatory reporting, and the restricted-use shipment microdata provide transaction-level geographic detail absent from public aggregate series. Section 4 returns to these institutions when motivating the model's assumptions about conduct.

2.4 From Shipment Links to Structural Procurement Markets

The model's product is a processing plant's fresh or frozen poultry, that is, a processing establishment crossed with fresh or frozen status; its market is a destination county in a survey year. Positive processor-origin shipments identify the plant-products available in that county. The shipment quantities then identify their local procurement shares. Firm identifiers in the Longitudinal Business Database (LBD) link plants under common ownership; I use those links to define each firm's set of plant-products in a county. Several physical suppliers may therefore represent fewer independent pricing decisions. In a county served by two plants of the same firm, substitution between those plants retains revenue within the firm; substitution to a rival-owned plant does not. Under the multiproduct Bertrand benchmark, each processor jointly prices its products across its plants while taking rival firms' prices as given.

The ZIP-level descriptive network and county-year structural markets serve successive parts of this construction. Destination ZIPs support channel classification and the measurement of sparse establishment-to-destination connections. County aggregation brings the suppliers serving a common procurement market into one choice set. Fresh and frozen forms distinguish products within a plant, while CMF product-line specialization assigns those products to broiler, processed-poultry, or turkey/other-poultry reference groups. This commodity grouping supplies the nests used in demand estimation; it does not replace the plant-by-temperature product definition.

Poultry products also differ in processing technology, temperature control, and delivery requirements. Fresh and chilled shipments differ from frozen shipments in inventory flexibility and delivery timing. As well as raising supply costs, distance can affect buyers' sourcing preferences through reliability and product quality, consistent with evidence that transit time acts as a trade barrier (Hummels and Schaur, 2013). The shipment data measure these

relationships through procurement prices, weights, and distances.

The poultry industry is organizationally distinctive. Broiler production has long relied on vertical coordination among growers, integrators, and processing establishments, and processing firms often operate multiple establishments (Martinez, 1999; Vukina, 2001; MacDonald, 2008). Concentration has also risen steadily; in Census of Manufactures data, the four-firm concentration ratio in broiler processing—the share of industry shipment value accounted for by the four largest firms—climbed from 44 percent in 1992 to 59 percent in 2007 (Weng, Vukina, and Zheng, 2015). In July 2022, a joint venture between Cargill and Continental Grain acquired Sanderson Farms and combined it with Continental Grain's Wayne Farms, forming Wayne-Sanderson Farms (Cargill, 2022). Poultry nevertheless remains the least concentrated of the major U.S. meat industries—four-firm ratios in beef packing exceed 80 percent—which makes it a setting where market power must be measured rather than presumed from structure alone. More broadly, modern agricultural markets often combine product differentiation, vertical coordination, and concentration, so market power cannot be evaluated from textbook assumptions alone (Sexton, 2013). These features imply that pricing incentives may depend on both plant-level cost conditions and firm-level ownership links. A multi-plant firm selling more than one poultry product into the same destination market internalizes some substitution across its own products when setting prices, while rival-owned plants do not.

The upstream grow-out stage, in which birds are raised to market weight, does not involve a spot market for live birds. Broiler production is organized through production contracts. As well as supplying chicks, feed, and veterinary services, integrators provide technical assistance, catching, and live-haul services; growers provide housing capital, labor, utilities, and day-to-day husbandry (MacDonald, 2014; Vukina, 2001). USDA survey evidence reports that 97 percent of broilers were raised on contract operations in 2011. The same evidence shows that grower–plant geography is highly local; the mean distance from a grower to the integrator's processing plant was 34 miles, and 90 percent of birds were produced within 60 miles of the plant (MacDonald, 2014). Because growers make long-lived, specialized investments in poultry houses and usually face only a few nearby integrators, upstream supply is better viewed as a governed contractual relationship than as an unconstrained spot market. This relationship is itself an active regulatory margin; recent Packers and Stockyards Act rulemaking has targeted tournament-payment disclosure and grower compensation (U.S. Department of Agriculture, Agricultural Marketing Service, 2023), and the Sanderson–Wayne transaction closed under a consent decree addressing grower pay (U.S. Department of Justice, 2022). Those instruments regulate the upstream boundary of the markets studied here; the downstream sales side, by the USDA's own account, has seen decades of little enforcement

attention (U.S. Department of Agriculture, Agricultural Marketing Service, 2024). This institutional structure motivates the plant-shutdown counterfactual in ten poultry-processing and production counties with a history of flooding, where a local disruption can interrupt nearby grower relationships and live-bird flows as well as downstream shipments. However, the simulation in Section 6 models only the downstream choice-set consequences.

3 Data

I construct the poultry procurement network by linking three objects: shipments from particular plants to destination markets, the firms that own those plants, and the downstream organizations whose purchasing decisions generate demand. The broad descriptive network contains approximately 73,000 shipment records across 2007, 2012, and 2017. It identifies shipment connections and channel organization. Aggregation and the estimation restrictions then produce plant-product quantities, prices, supplier choice sets, and local procurement shares for a demand-and-supply model. Appendix B gives the network statistics, geographic comparison, market-size construction, and representative-buyer calibration.

3.1 CFS Shipments and Census Microdata

The Commodity Flow Survey (CFS) is a shipper-based survey conducted jointly by the U.S. Census Bureau and the Bureau of Transportation Statistics. The survey is conducted every five years and provides repeated cross-sectional information on freight shipments originating from selected business establishments. The sampling frame covers establishments in manufacturing, mining, wholesale trade, and selected retail and service industries, together with auxiliary establishments such as warehouses and distribution centers, while excluding several important categories such as farm-origin shipments and imports. Sampled establishments report outbound shipments for assigned one-week periods in each calendar quarter of the survey year, allowing the CFS to capture seasonal variation in freight flows. For each reported shipment, the CFS records commodity type, shipment value, weight, transportation mode, origin, destination, distance shipped, and related shipment characteristics (U.S. Census Bureau, 2025, 2017).

I focus on poultry-related commodities and conduct the analyses using the 2007, 2012, and 2017 waves of the CFS. Poultry shipments are identified using the Standard Classification of Transported Goods (SCTG) commodity codes for fresh or chilled poultry and frozen poultry; live-poultry shipments do not enter the descriptive network or the estimation sample. I use these repeated cross-sections to construct poultry shipment networks and compare the

geographic structure of poultry flows across survey years.

The CFS reports two distinct weight concepts: the physical shipment weight reported for shipment record s from establishment e, w_{se}^{ship}, and the within-establishment annualization weight used in this paper, $\kappa_{se}^{\text{plant}}$. The latter is the product of five CFS component weights,

$$\kappa_{se}^{\text{plant}} = \kappa_{se}^{\text{ship}} \kappa_{se}^{\text{shipNR}} \kappa_{se}^{\text{quarter}} \kappa_{e}^{\text{quarterNR}} \kappa_{e}^{\text{estAdj}}.$$

The first four components expand usable records to shipments made by the same establishment during its reporting week, quarter, and survey year; the fifth calibrates that establishment's annual estimate to its CFS stratification measure. I do not apply an establishment first-stage sample weight or a nonresponse post-stratification adjustment, either of which would make responding establishments represent other establishments. Thus, the expanded quantities retain the identity of the observed establishment. I use "CFS-expanded shipment weight" to mean $\tilde{w}_{se} = \kappa_{se}^{\text{plant}} w_{se}^{\text{ship}}$. Similarly, v_{se} is the reported shipment value, and $\tilde{v}_{se} = \kappa_{se}^{\text{plant}} v_{se}$ is the CFS-expanded shipment value. These expanded weights and values are the building blocks for product-market quantities, procurement prices, weighted distances, and channel shares below.

I link the CFS shipment records to establishment-level businesses through the Longitudinal Business Database (LBD). The LBD is an annual, longitudinal establishment database constructed from the Census Bureau's Business Register and related administrative records. It covers private, nonfarm employer establishments in the United States and provides longitudinal establishment identifiers, firm identifiers, establishment location, employment, payroll, and industry classification using the North American Industry Classification System (NAICS) (Jarmin and Miranda, 2002). These NAICS codes help identify processors, wholesalers, retailers, and other downstream establishment types.

I use the longitudinal establishment identifiers to follow plants over time and connect CFS shipment records to plant-level characteristics. Firm identifiers group establishments under common ownership and distinguish single-plant firms from multi-establishment firms. For manufacturing establishments, I further merge plant-level information from the Census of Manufactures (CMF) and the Annual Survey of Manufactures (ASM), which provide measures of employment, payroll, inventories, assets, capital expenditures, materials costs, fuel and electricity expenditures, and detailed product-line shipments. Together, these linked datasets characterize poultry shipments by plant scale, input use, cost shifters, product specialization, and firm ownership structure. They also identify when multiple products or shipments in the same destination market are supplied by establishments under common ownership, which is central for modeling multi-product pricing incentives.

On the demand side, I use LBD establishments within 100 miles of each destination-county-year market to characterize the local food-service, retail, and wholesale procurement decision-makers. Later subsections use these records to measure downstream buyer composition and to help distinguish retail-oriented from wholesale-oriented shipment destinations. I also use county-level wholesale and retail activity from the Annual Wholesale Trade Survey (AWTS) and Annual Retail Trade Survey (ARTS) as auxiliary information on local downstream market capacity (U.S. Census Bureau, 2024a,b). These linked shipment, establishment, and firm records support a demand-and-supply framework with endogenous procurement prices, plant-level marginal-cost heterogeneity, multi-product ownership, and market-specific downstream buyer composition.

3.2 Classifying Downstream Shipment Channels

CFS identifies a shipment's destination ZIP code but does not identify its receiving establishment. I infer the procurement channel by linking destination ZIPs to candidate recipient establishments in the LBD. The retail group contains retail stores and retail-owned warehouses; the wholesale group contains food wholesalers and wholesale-owned warehouses. A ZIP containing only the retail group supplies a processor-to-retailer (PR) training label, while a ZIP containing only the wholesale group supplies a processor-to-wholesaler (PW) training label; PR shipments primarily carry household-serving procurement and PW shipments primarily carry business-serving procurement. The current labeled sample contains approximately 17,000 shipments, with disclosure-rounded counts of 15,580 PR and 1,300 PW records. Shipments to ZIPs with both groups enter the unlabeled prediction sample. Appendix B.1 gives the establishment and owner-industry rules.

I partition the labeled records in a ratio of 90 to 10 into an initial training set and a validation set. Ten-fold cross-validation on the training portion compares random-forest specifications (Breiman, 2001), and the ten-percent validation portion is used during model selection. The predictors are shipment weight, value, distance, commodity and fresh/frozen status, transport mode, year, month, day of week, and supplying-plant characteristics and firm affiliation. Destination demographic variables are excluded. The labeled and mixed-destination ZIP groups do not differ significantly in population, income, education, or area. After specification selection, I refit the classifier on all labeled shipments and predict the channels of mixed-destination shipments.

The ten-percent validation sample yields about 96 percent accuracy. The confusion matrix in Table A1 instead reports fitted classifications after the final refit on the full labeled sample. Its columns are reference channels and its rows are predicted channels: 1,200 PW

and 15,500 PR records lie on the diagonal, 100 PW records are classified as PR, and 80 PR records are classified as PW. These rounded fitted counts and the held-out validation statistic refer to different stages of the procedure. Channel assignment is a supervised measurement step before structural estimation (Mullainathan and Spiess, 2017). It classifies the processor's procurement channel rather than recovering a named receiver or the final consumer. Wholesaler-origin shipments receive the descriptive label *From W*.

For product j in market m, the wholesale-channel share is

$$ToW_{jm} = \frac{\sum_{s\in(j,m)} \tilde{w}_{se}\mathbf{1}\{s \in PW\}}{\sum_{s\in(j,m)} \tilde{w}_{se}}, \qquad ToR_{jm} = 1 - ToW_{jm}. \tag{1}$$

The denominator includes all processor-origin shipments in the cell. The demand specification uses the retail share ToR_{jm} in levels and interacted with log distance. The supply specification retains wholesale-channel measures. Both specifications include measures of the channel composition of a particular plant-product's shipments into a market.

A separate market-level measure summarizes business-demand orientation in the baseline equilibrium:

$$PWShare_m^{\text{base}} = \frac{\sum_{j\in\mathcal{J}_m} q_{jm}^{\text{base}} ToW_{jm}}{\sum_{j\in\mathcal{J}_m} q_{jm}^{\text{base}}}. \tag{2}$$

This quantity-weighted county measure is the baseline wholesale procurement share, or PW share, used in the post-estimation county comparisons of market scale and counterfactual responses. It differs from the cell-level ToW_{jm} and from the representative buyer-type probabilities below. A wholesale receipt can reflect either a wholesaler's own sourcing decision or a food-service organization's order delivered through a wholesaler.

3.3 The Shipment Network and Channel Organization

For each channel and survey year, a directed edge connects a shipping establishment to a destination ZIP when the shipment records contain a positive flow between them. I retain the establishment identifier at the origin, which preserves the distinction between a plant and its firm. This establishment–ZIP network provides the channel statistics in Table 1. The table reports averages of annual means and sums of annual counts across 2007, 2012, and 2017. Its mean density averages the three annual network densities rather than recomputing a density from the summed counts.

Processor-to-wholesaler shipments average 90,169 pounds and 636 miles, compared with 25,985 pounds and 430 miles for processor-to-retailer shipments. Wholesale-bound processor shipments are therefore about 3.5 times as large and travel 48 percent farther. Their mean

Table 1: The Scale, Reach, and Sparsity of Poultry Procurement Channels

	PR	PW	*From W*
Panel A. Means across survey years			
Price ($/lb)	1.39	1.45	1.52
Shipment weight (lb)	25,985	90,169	18,345
Shipment distance (miles)	430	636	83
Outbound destination HHI	0.24	0.64	0.22
Network density (county level)	10.0%	14.0%	12.0%
Panel B. Counts summed across survey years			
Source establishments	600	440	300
Destination ZIP codes	4,500	1,000	3,600
Shipment records	44,000	11,600	17,200
Establishment–destination edges	28,000	7,000	13,000

Notes: PR denotes processor-to-retailer shipments; PW denotes processor-to-wholesaler shipments; *From W* denotes wholesaler-origin shipments. Panel A averages the annual summary measures for 2007, 2012, and 2017. Panel B sums annual counts, so establishments, ZIP codes, and edges may recur across years. HHI first measures each source establishment's concentration of outbound weight across destination ZIP codes, then averages establishments using outbound shipment weight. Annual density divides distinct establishment–county edges by the product of source-establishment and destination-county counts; the displayed value is the mean of annual densities. The other destination statistics and edge counts use ZIP codes. Appendix B.2 gives the formulas.

prices are respectively $1.45 and $1.39 per pound. PW reaches fewer destination ZIPs and has a higher outbound destination HHI, 0.64 versus 0.24. *From W* shipments average 18,345 pounds and 83 miles at $1.52 per pound, with destination HHI 0.22. The organization is consistent with large, long-distance purchases being assembled through wholesalers and smaller, shorter shipments serving downstream demand. This comparison combines channel-level shipment records rather than tracing individual lots through every stage.

The HHI measures how a shipping establishment's pounds are concentrated across destination ZIPs and then averages those establishment HHIs using shipment-volume weights. The inverse of each reported channel-average HHI provides an effective-destination benchmark: about four ZIPs for processor-to-retailer shipments and 1.6 for processor-to-wholesaler shipments. This benchmark describes outbound geographic concentration, not the average number of destinations per establishment; the structural model instead studies the firms competing within a destination market. The network is sparse; mean county-level network densities of 10.0 percent for PR, 14.0 percent for PW, and 12.0 percent for *From W* show that observed links occupy a small share of possible establishment–county pairs. Appendix B.2 gives both definitions and the cross-year aggregation.

The shipment links determine more than the geography of aggregate demand. After ZIP destinations are assigned to counties, positive processor-origin flows define the local product

set, and expanded quantities identify the shares of those suppliers. Linking the establishments to firms then identifies which products are jointly priced. This sequence—shipment network, local supplier choice sets, and local ownership structure—connects the descriptive data to the model.

3.4 Products, Markets, Prices, and Quantities

The structural estimation retains shipments with SCTG code 05121 (poultry fresh or chilled) and SCTG code 05122 (poultry frozen). A product j is a plant-product: an establishment interacted with one of these two SCTG codes, equivalently a processing establishment crossed with fresh/chilled or frozen status. A market is a destination-county-year cell, indexed by m, with choice set $\mathcal{J}_m$. This definition treats a plant shipping fresh poultry and the same plant shipping frozen poultry as two separate products, while preserving the plant identity needed for ownership and cost controls. Only processor-origin shipments enter the estimation sample; wholesaler-origin (*From W*) shipments appear in the descriptive network but not in the structural choice set.

Physical shipment weight and the within-establishment annualization weight are distinct variables. For each shipment record, I first multiply the reported shipment weight by $\kappa_{se}^{\text{plant}}$ and then aggregate within product-market cells. Thus, product-market quantity, q_{jm}, is measured as expanded shipment weight,

$$q_{jm} = \sum_{s \in (j,m)} \tilde{w}_{se} = \sum_{s \in (j,m)} \kappa_{se}^{\text{plant}} w_{se}^{\text{ship}}. \tag{3}$$

As a scale check, I compare the within-establishment annualized CFS shipment total $\sum_s \kappa_{se}^{\text{plant}} w_{se}^{\text{ship}}$ for the retained poultry scope with the USDA production benchmark for the matched survey years and poultry classes. The retained expanded CFS total is about 85 percent of that benchmark (USDA NASS, various years). This comparison is used only to assess scale; the model quantities remain annualized shipments from observed establishments, not a complete production census.

The CFS instructs respondents to report shipment value net of freight charges and excise taxes, on a free-on-board (FOB) plant basis (U.S. Census Bureau, 2017). Accordingly, the product-market procurement price is the expanded-value-to-expanded-weight ratio,

$$p_{jm} = \frac{\sum_{s \in (j,m)} \tilde{v}_{se}}{\sum_{s \in (j,m)} \tilde{w}_{se}}. \tag{4}$$

Equivalently, this is $\sum_s \kappa_{se}^{\text{plant}} v_{se} / \sum_s \kappa_{se}^{\text{plant}} w_{se}^{\text{ship}}$ within the product-market cell. I refer to p_{jm}

as the procurement price. The procurement price measures the supplying plant's shipment value per pound and excludes separately billed freight charges. The within-establishment annualization weights enter at the shipment level before aggregation, and all monetary variables are converted to constant 2017 dollars before estimation. Shipment distance is aggregated with the same expanded shipment weights,

$$d_{jm} = \frac{\sum_{s \in (j,m)} \text{distance}_s \times \tilde{w}_{se}}{\sum_{s \in (j,m)} \tilde{w}_{se}}. \tag{5}$$

The estimation uses $\log(d_{jm})$.

Plant-level product categories are constructed from CMF product-line information rather than from shipment temperature status. The CMF reports establishment output using detailed NAICS product codes. For poultry processing, these codes distinguish young chickens; processed poultry and small-game products containing at least 20 percent poultry or meat; turkeys; hens and fowl; and other poultry and small game (Weng, Vukina, and Zheng, 2015). In my sample, poultry establishments are highly specialized, so the detailed CMF product code uniquely identifies each establishment's underlying product category. I assign each establishment to one mutually exclusive plant-level category: broiler, processed poultry, or the turkey/other-poultry reference group. These CMF categories supply the commodity nests and the broiler and processed-poultry indicators in demand and supply; the reference group retains the existing other-poultry classification, including turkey. An establishment can still ship both fresh or chilled and frozen forms of its underlying product. The SCTG code determines only the frozen indicator: code 05122 is frozen and code 05121 is fresh or chilled. Additional plant-level controls include production employment, asset stock, hourly production-worker pay, capital expenditure rate, materials unit cost, supplemental input unit cost, labor unit cost, inventory-to-assets ratio, and indicators for own-slaughter and external-slaughter processing plants. The three unit-cost variables are reported in constant 2017 dollars per pound. Appendix C provides variable definitions.

3.5 Market Size and Shares

Market size converts CFS-expanded inside shipments into product shares and an outside option. Four objects enter the demand model: observed inside shipments R_m, final market size M_m, product shares s_{jm}, and the outside share s_{0m}. The construction starts from observed CFS inside shipments, uses county population and national poultry availability to form a baseline capacity measure, and then adjusts that scale using downstream retail and wholesale activity from ARTS/AWTS and LBD. Appendix B.5 gives the exact market-size formula and

small-area construction.

Observed inside shipments in destination-county-year market m are

$$R_m = \sum_{j \in \mathcal{J}_m} q_{jm}. \tag{6}$$

Here, R_m is the total CFS-expanded quantity of retained poultry products shipped into market m. This is not total county poultry consumption and not total poultry sales by all suppliers. I use R_m as the dependent variable in the auxiliary market-size regression and as a lower bound on final market size. The fitted market-size proxy should therefore be read as a capacity measure for potential poultry procurement in a destination-county-year market, not as a census of realized poultry consumption. Product and outside shares are

$$s_{jm} = \frac{q_{jm}}{M_m}, \qquad s_{0m} = 1 - \sum_{j \in \mathcal{J}_m} s_{jm}. \tag{7}$$

The outside option is a residual normalization rather than an observed choice. It can be interpreted as covering procurement from establishments or sourcing arrangements not included in the estimation sample, including shipments excluded by the sample restrictions in Subsection 3.7 and substitution toward other meat products, although these flows are not separately observed or modeled. Because the market-size proxy is calibrated to downstream meat retail and wholesale capacity rather than to poultry shipments alone, it deliberately leaves room for this residual, and the counterfactual welfare comparisons hold the resulting normalization fixed.

3.6 Market-Specific Procurement Decision-Makers

Retailers, wholesalers, and food-service organizations can differ in procurement scale and price sensitivity. The buyer type in the demand model identifies the organization choosing the upstream supplier. A food-service company may choose a processor while a wholesaler handles delivery; a wholesaler may instead source on its own account. Both arrangements can produce a PW shipment. Household- versus business-serving channel orientation therefore does not reduce the model to two buyer types.

For each destination-county-year market m, I identify eligible LBD establishments within 100 miles and classify them into food service (F), retail (R), and wholesale (W). Let E_{gm} denote employment in type g within this local area. National throughput-per-employee benchmarks, normalized to food service, are 1, 2.9, and 38. I apply a 60:40 benchmark for food-service-led versus wholesaler-led procurement control, with retail unchanged. The

resulting adjusted factors are $a_F = 0.6$, $a_R = 2.9$, and $a_W = 15.2$. Local employment then determines the market-specific probability of a procurement decision of each type:

$$\phi_{gm} \equiv \Pr(g \mid m) = \frac{a_g E_{gm}}{0.6E_{Fm} + 2.9E_{Rm} + 15.2E_{Wm}}, \qquad g \in \{F, R, W\}. \tag{8}$$

The 60:40 benchmark adjusts procurement control, rather than fixing a county's buyer shares or its observed PW share. Markets with different food-service, retail, and wholesale employment have different normalized probabilities.

I draw 1,000 representative procurement decision nodes with replacement for each market, weighting each eligible establishment by employment times its type's adjusted factor. The empirical distribution of these draws approximates ϕ_{gm}. Each node retains a buyer-type indicator and the within-type taste draw used in the model. Simulated product shares average conditional choice probabilities over the 1,000 nodes. Appendix B.6 provides the industry mapping, throughput benchmarks, and sampling probabilities.

The retail and wholesale indicators are

$$D^R_{im} = \mathbf{1}\{T_{im} = R\}, \qquad D^W_{im} = \mathbf{1}\{T_{im} = W\}, \tag{9}$$

with food service as the omitted type. They allow price sensitivity to differ by procurement decision-maker. Employment affects the sampling probabilities, rather than entering again as an individual demand characteristic. The simulated buyer probabilities, the observed cell-level channel shares, and the quantity-weighted baseline county PW share thus have distinct roles: integration over purchasing decisions, shipment-route composition, and a summary of baseline business-demand orientation.

3.7 Sample Restrictions and Descriptive Statistics

The estimation sample excludes establishments with internal plant transfers greater than 30 percent of total shipment value, trims product-market observations outside the 5th and 95th percentiles of the procurement-price distribution, drops product-market observations with shares below 0.1 percent, and removes markets with fewer than four inside products. Shipment volume excluded by these restrictions is not reassigned to retained products; under the market-size construction in Subsection 3.5, it falls into the residual outside option as a matter of normalization rather than as an observed reallocation. The final sample contains 700 destination-county-year markets, 6,500 product-market observations, 400 plant-products, and 700,000 representative buyer records.

Table A2 reports descriptive statistics for the estimation variables. The average procure-

ment price is 1.423 dollars per pound ($/lb), and the average product share is 0.046. The mean log distance is 6.09, with substantial dispersion across product-market cells. The sample is concentrated in broiler products but still contains meaningful frozen and processed-poultry variation; broiler, frozen, and processed-poultry indicators have means of 0.600, 0.277, and 0.246, respectively. The mean cell-level wholesale-channel share ToW is 0.093, with a standard deviation of 0.282. This simple mean over retained product-market observations differs from both the quantity-weighted county PW share in (2) and the buyer-type probabilities in (8). These statistics document heterogeneity in the retained estimation cells.

Because the CFS is a sample of shipments rather than a census, each product-market cell aggregates a finite number of sampled shipment records, and the choice set in each market is defined by sampled positive flows. Table A3 reports the resulting sampling depth. Thin cells are common but economically small; cells built from one or two sampled shipments account for 29.5 percent of observations but only 4.5 percent of CFS-expanded shipment weight, while cells with five or more shipments carry 83 percent of expanded weight, and the weight-weighted median cell is built from about 12 shipments. Thus, sampling noise in shares and prices is concentrated where the welfare stakes are smallest.

I assess destination-set coverage directly using a plant-level occupancy diagnostic. For each plant–survey-year observation, I use the number of sampled shipment records and the number of distinct destination counties among those records to estimate the plant's potential county destinations under an equal-probability occupancy model. The disclosure-rounded average ratio of observed counties to estimated potential counties is about 95 percent. Thus, county aggregation appears to recover nearly all of the effective plant-level destination support implied by this benchmark; Appendix B.4 gives the estimator and its assumptions. The diagnostic cannot rule out rare, low-probability links, and the cells retained after the share restrictions are still selected partly on realized demand, as emphasized by Gandhi, Lu, and Shi (2023). The plant–county availability component of the positive-flow choice set is therefore well covered empirically, without implying that every technically feasible link is observed. Unsampled and excluded flows fall into the residual outside option as a matter of normalization (Subsection 3.5), and the estimates should be read as conditional on the retained choice sets.

4 Model and Estimation

The shipment network supplies three objects needed to study local competition: the plant-products available in each destination market, their procurement shares, and the firms that set their prices. The buyer distribution constructed in Section 3 describes whose sourcing

decisions those suppliers face. I combine these objects in a random-coefficients nested-logit demand system and a multiproduct Bertrand pricing model. Demand determines substitution among local sourcing options; plant-to-firm links determine which of those substitutions a processor retains within the firm. Together, they identify price–cost margins and the responses to supplier shutdowns, higher costs, and consolidation.

The approach follows the differentiated-products demand framework of Berry (1994) and Berry, Levinsohn, and Pakes (1995), its application to supply-side markup estimation by Nevo (2001), and the implementation guidance in Nevo (2000a), Dubé, Fox, and Su (2012) and Conlon and Gortmaker (2020). Local buyer information enters through the market-specific distribution over procurement decision-makers. This use of local composition complements the identification from aggregate market shares (Berry and Haile, 2014, 2024; Conlon and Gortmaker, 2025).

4.1 Demand

A product j is a processing plant's fresh or frozen poultry, that is, a processing establishment crossed with fresh or frozen status, and market m is a destination county in a survey year. Plants in the sample specialize in one commodity category: broiler, processed poultry, or turkey and other poultry, measured from CMF product lines (Subsection 3.4). That category determines the product's nest separately from its temperature status. Thus, a plant's fresh and frozen products retain the same plant identity and firm affiliation while constituting distinct sourcing options.

Buyer types represent control of the upstream sourcing decision. As noted, a food-service company can select a processor while a wholesaler receives and distributes its shipment, and a wholesaler can also select the processor on its own account. The first decision is food-service-led and the second is wholesaler-led, even when both shipments use the PW channel. The LBD distribution assigns weights to food-service, retail, and wholesale decision-makers based on local employment within 100 miles and adjusted throughput factors of 0.6, 2.9, and 15.2. The 60/40 procurement-control benchmark adjusts national throughput measures, while local employment determines each market's resulting mix. I draw 1,000 representative procurement decisions per market from this distribution, as detailed in Subsection 3.6.

For representative buyer i, let $f(j,m)$ denote product j's LBD firm, $o(j)$ its plant's origin state, $r(m)$ the destination state, and $t(m)$ the survey year. Indirect utility is

$$u_{ijm} = \delta_{jm} + \mu_{ijm} + \varepsilon_{ijm}, \tag{10}$$

where ε_{ijm} has the nested extreme-value structure. Mean utility is

$$\delta_{jm} = x'_{jm}\beta + \phi_{f(j,m)} + \lambda_{o(j),t(m)} + \eta_{r(m),t(m)} + \xi_{jm}, \tag{11}$$

where x_{jm} collects observed product-market demand characteristics, β is the corresponding vector of mean-utility coefficients, $\phi_{f(j,m)}$ is a firm fixed effect, $\lambda_{o(j),t(m)}$ is an origin-state-by-survey-year fixed effect, $\eta_{r(m),t(m)}$ is a destination-state-by-survey-year fixed effect, and ξ_{jm} is unobserved demand quality. The buyer-specific component is

$$\mu_{ijm} = -\alpha_{im} p_{jm}. \tag{12}$$

Here, p_{jm} is the shipment-based price in dollars per pound, defined in Subsection 3.4. The price-sensitivity parameter is positive and lognormally distributed conditional on buyer type:

$$\log \alpha_{im} = \pi_F + \pi_R D^R_{im} + \pi_W D^W_{im} + \sigma v_{im}, \qquad v_{im} \sim \mathcal{N}(0,1). \tag{13}$$

The indicators D^R_{im} and D^W_{im} identify retail and wholesale decision-makers. Food service is the reference type; π_F is its baseline log price sensitivity, and π_R and π_W are the retail and wholesale increments. Exponentiation makes α_{im} positive, so price lowers utility for every buyer. The common dispersion parameter σ allows variation within each type. These differences in price response, combined with the local buyer mix, determine the curvature of market demand and its role in cost pass-through and merger effects (Miravete, Seim, and Thurk, 2023).

Products are partitioned by their plant's CMF commodity classification into broiler, processed-poultry, and a turkey/other-poultry reference group:

$$g(j) \in \mathcal{G} = \{B, P, O\}. \tag{14}$$

The frozen indicator remains an observed product characteristic in mean utility rather than a nest definition. Let $\mathcal{J}_{g,m}$ denote products in nest g in market m, and let $\rho \in [0,1)$ be the nesting parameter. With the outside option normalized to zero, define the agent-specific inclusive value as

$$I_{igm}(\theta) = \sum_{k \in \mathcal{J}_{g,m}} \exp\left(\frac{\delta_{km} + \mu_{ikm}}{1-\rho}\right). \tag{15}$$

Agent i's probability of choosing product j is the product of the conditional within-nest

probability and the nest probability:

$$P_{ij|g(j),m}(\theta) = \frac{\exp\left((\delta_{jm} + \mu_{ijm})/(1-\rho)\right)}{I_{ig(j),m}(\theta)}, \tag{16}$$

$$P_{ig,m}(\theta) = \frac{I_{igm}(\theta)^{1-\rho}}{1 + \sum_{h \in \mathcal{G}} I_{ihm}(\theta)^{1-\rho}}, \tag{17}$$

$$P_{ijm}(\theta) = P_{ij|g(j),m}(\theta) P_{ig(j),m}(\theta). \tag{18}$$

When $\rho = 0$, these probabilities collapse to standard random-coefficients logit probabilities. Positive ρ allows closer substitution within each commodity nest. Model-implied product shares average the conditional probabilities over the market's integration nodes:

$$s_{jm}(\theta) = \sum_{i=1}^{1{,}000} \psi_{im} P_{ijm}(\theta), \qquad \psi_{im} = \frac{1}{1{,}000}. \tag{19}$$

The sampling probabilities already incorporate employment, throughput, and procurement control, so each sampled node receives the equal integration weight ψ_{im}. Observed shares are inverted into mean utilities using the BLP contraction mapping (Berry, 1994; Berry, Levinsohn, and Pakes, 1995).

The vector x_{jm} contains frozen status, broiler and processed-poultry indicators, private-carriage share, retail procurement share ToR_{jm}, log shipment distance, and distance interactions with the product and shipment attributes. The turkey/other-poultry group is the reference commodity category. Retail share enters both in levels and interacted with log distance, allowing the value of proximity to vary with the household-serving component of procurement. This product-market channel variable is distinct from the buyer-type indicators in (13) and from the county-level baseline wholesale procurement share (PW share) used in the post-estimation county comparisons below, which is not a model parameter.

The three fixed-effect sets absorb persistent firm-level quality, common origin-state-by-year shocks, and common destination-state-by-year demand conditions. Standalone survey-year effects are absorbed by the state-by-year effects. Remaining product-market demand variation is represented by ξ_{jm}. The resulting specification separates buyer-specific price sensitivity, commodity substitution, and the geographic attractiveness of a supplier.

4.2 Supply

Each processor jointly prices its products across its plants, taking rival processors' prices as given. Let $\mathcal{J}_{fm}$ be firm f's products in market m. Its operating surplus is

$$\Pi_{fm} = M_m \sum_{j \in \mathcal{J}_{fm}} (p_{jm} - mc_{jm}) s_{jm}(\mathbf{p}_m; \theta), \tag{20}$$

where M_m is calibrated market size. The same-firm indicator $\mathcal{H}_{jk,m}$ equals one when products j and k belong to the same LBD firm and zero otherwise. Under multiproduct Bertrand–Nash competition, the pricing conditions are

$$\mathbf{p}_m = \mathbf{mc}_m + \Delta_m^{-1} s_m(\theta), \tag{21}$$

where

$$\Delta_{jk,m} = -\mathcal{H}_{jk,m} \frac{\partial s_{km}(\theta)}{\partial p_{jm}}. \tag{22}$$

The implied markup is $p_{jm} - mc_{jm}$. If a buyer switches from one plant-product to another belonging to the same firm, the firm retains the sale. The same-firm indicator lets pricing account for these retained sales. Consequently, two markets with similar numbers of supplying plants can have different pricing incentives when those plants belong to different sets of firms. This is the economic role of linking the shipment network to LBD firm identifiers.

Marginal cost is parameterized in levels with a linear regression,

$$mc_{jm} = w_{jm}' \gamma + \omega_{jm}, \tag{23}$$

where w_{jm} includes plant-level unit costs and scale, input-mix variables, product and shipment characteristics (including log shipment distance, plant-type and year indicators), destination population, and rival-count controls. The supply equation retains the wholesale-channel share ToW_{jm} and its distance interaction. Marginal cost is measured in dollars per pound, and ω_{jm} is the unobserved cost component. The three fixed-effect sets specified for demand are not imposed on costs. Demand and cost parameters are estimated jointly through the moments below.

The annual pricing benchmark reflects the recurring nature of poultry procurement. Quotation-linked agreements and periodic renegotiation allow transaction prices to respond to market conditions within the survey year, as described in Subsection 2.3. The model summarizes this annual response through firms' prices across their plant-products (Nevo, 2001; Miller and Osborne, 2014). The resulting margins measure the incentives generated by multiproduct competition, and the counterfactuals hold this conduct specification fixed while

changing availability, costs, or the firms that price the products.

4.3 Instruments, Identification, and Estimation

Price is endogenous because unobserved supplier quality and buyer–supplier matching can affect both prices and procurement shares. Excluded demand instruments combine plant cost and scale measures, rival product and firm counts, and local differentiation measures. Plant measures include unit input costs, production employment, assets, and production structure. The Gandhi–Houde differentiation instruments measure the proximity of rival products in the space of shipment distance, frozen status, and commodity characteristics (Gandhi and Houde, 2019). These sources of variation connect price to production conditions and the competitive environment, while the included demand characteristics and fixed effects control for observed supplier and market attributes. Appendix B.7 lists the instruments.

The product's own log shipment distance is an included demand characteristic and serves as its own instrument. Distance-based differentiation measures are separate excluded instruments for price. Identification requires orthogonality between the demand instruments and unobserved demand quality after accounting for the included characteristics and the three demand fixed-effect sets. I use a fixed instrument set, following the distinction between instrument choice and demand specification emphasized by Reynaert and Verboven (2014).

Demand and supply contribute the moment conditions

$$\mathbb{E}[z_{jm}^{d}\xi_{jm}(\theta)] = 0, \tag{24}$$

$$\mathbb{E}[w_{jm}\omega_{jm}(\theta,\gamma)] = 0. \tag{25}$$

Here, z_{jm}^{d} contains the demand instruments and w_{jm} contains the included cost regressors from (23). The supply moments fit recovered marginal costs to those regressors; there is no separate set of excluded supply-side instruments. The vector $\theta = (\beta', \pi_F, \pi_R, \pi_W, \sigma, \rho)'$ collects the demand parameters, and γ collects the cost coefficients. The cost residual is evaluated using marginal costs recovered from the pricing conditions, so demand and supply are linked within estimation. Firm, origin-state-by-year, and destination-state-by-year fixed effects are absorbed from the demand regressors and instruments.

The parameters are estimated by two-step GMM (Hansen, 1982),

$$(\hat{\theta}, \hat{\gamma}) = \arg\min_{\theta,\gamma} \; \hat{g}(\theta,\gamma)'\Omega\hat{g}(\theta,\gamma), \tag{26}$$

where $\hat{g}$ stacks the demand and cost sample moments and Ω is the GMM weighting matrix. The implementation uses L-BFGS-B optimization and `pyblp` routines for estimation and

post-estimation analysis (Conlon and Gortmaker, 2020). Each of the 1,000 integration nodes per market contains a buyer-type indicator and a standard-normal draw for within-type log price sensitivity. Standard errors are clustered by destination county, allowing dependence across products and survey years within the same county.

The preliminary IV logit has a fixed-effect-adjusted Montiel Olea–Pflueger effective first-stage F statistic of 77.62 (Montiel Olea and Pflueger, 2013). For the nested-logit benchmark, which instruments both price and the within-nest share, the Kleibergen–Paap rk Wald F statistic (where rk denotes rank) is 58.19 (Kleibergen and Paap, 2006). The first-stage projections absorb the same three fixed-effect sets as demand. Section 5 reports the associated price and substitution estimates.

5 Results

The results connect the local supplier sets identified by the network to three economic objects: buyer substitution, processor margins, and the scale of regional procurement. I first establish differences in price sensitivity and the value of proximity, then use the pricing conditions to recover market power. The final comparison relates baseline county procurement and gains from trade to business-demand orientation, proxied by the wholesale procurement share (PW share).

5.1 Preliminary Logit Benchmarks

The preliminary estimates in Table 2 motivate the treatment of price endogeneity and commodity substitution in the structural model. Instrumenting price moves its coefficient from -1.062 to -1.898. This movement is consistent with favorable unobserved demand conditions raising both prices and purchases in the simple logit. The nested-logit estimate of $\rho = 0.448$ indicates closer substitution within poultry categories. Baseline log distance is negative in all three specifications, while the broiler interaction offsets much of the category component. Appendix F gives the benchmark equations.

5.2 Buyer Heterogeneity and Procurement Organization

Table 3 reports the price-sensitivity parameters. Food service is the reference procurement decision-maker, with baseline log sensitivity $\pi_F = 1.1380$. The retail increment is $\pi_R = 0.6862$ and the wholesale increment is $\pi_W = 1.2020$. With a common within-type dispersion, exponentiating these increments gives price-sensitivity ratios of 1.99 for retail and 3.33 for wholesale relative to food service. The reported 95 percent confidence intervals are

Table 2: Preliminary Demand Estimates with Fixed Effects

Variable	Logit	IV logit	Nested logit
Price	-1.0620***	-1.8980***	-2.7150***
	(0.0823)	(0.2929)	(0.2821)
Log shipment distance	-0.2131***	-0.1851***	-0.1956***
	(0.0469)	(0.0466)	(0.0453)
Broiler × log distance	0.1839***	0.1550***	0.2184***
	(0.0491)	(0.0491)	(0.0475)
Nesting parameter (ρ)	–	–	0.4480***
	–	–	(0.0216)
Firm fixed effects	Yes	Yes	Yes
Origin-state-by-year fixed effects	Yes	Yes	Yes
Destination-state-by-year fixed effects	Yes	Yes	Yes
Observations	6,500	6,500	6,500

Notes: Selected coefficients from the preliminary demand specifications. Standard errors clustered by destination county are in parentheses. Each column uses the same product-market sample and absorbs LBD firm, origin-state-by-survey-year, and destination-state-by-survey-year fixed effects. The IV logit instruments price; the nested logit also instruments the within-nest share. The corresponding fixed-effect-adjusted first-stage statistics are 77.62 and 58.19. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

$[1.29, 3.05]$ and $[2.68, 4.13]$, respectively. Wholesale price sensitivity is about 1.67 times retail sensitivity. The dispersion estimate, $\sigma = 0.3644$, also permits substantial differences among decision-makers of the same type.

These estimates distinguish the organization of procurement from its physical delivery channel. Retailers primarily organize household grocery demand, while wholesalers primarily organize business and institutional procurement. Among these demand aggregators, the larger wholesale coefficient indicates a stronger response to supplier prices. Its difference from the food-service coefficient is also consistent with more price-oriented sourcing when many business purchases are organized into one procurement decision. Food-service-led orders can also be delivered through wholesalers. Market demand therefore combines all three decision-maker types, with the local weights in Appendix B.6, their within-type variation, and the available supplier set determining the aggregate response.

5.3 Distance, Product Differentiation, and Marginal Cost

Distance enters into both the attractiveness of a supplier and the cost of serving a market. With regard to demand, the retail-share interaction with log distance is -0.1433, with a standard error of 0.0766 (Table A4). Holding prices and other included characteristics fixed,

Table 3: Random-Coefficients Price Sensitivity by Buyer Type

Parameter	Interpretation	Estimate
π_F	Food-service baseline log price sensitivity	1.1380^{***}
		(0.1244)
π_R	Retail increment relative to food service	0.6862^{***}
		(0.2195)
π_W	Wholesale increment relative to food service	1.2020^{***}
		(0.1109)
σ	Within-type lognormal dispersion	0.3644^{***}
		(0.0441)

Notes: The estimation sample contains 6,500 product-market observations in 700 destination-county-year markets with 400 plant-products. Standard errors clustered by destination county are in parentheses. Demand absorbs firm, origin-state-by-survey-year, and destination-state-by-survey-year fixed effects. Food service is the reference buyer type in (13). The retail and wholesale increments shift log price sensitivity; exponentiation gives the corresponding type ratios. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

procurement with a larger household-serving retail component places a stronger penalty on distant suppliers. This adds a geographic dimension to the differences in price sensitivity; buyer composition matters for both the willingness to switch suppliers and the value of proximity. The retail-share level coefficient is -1.433, with a standard error of 0.4908.

Product categories contribute a second source of distance heterogeneity. The turkey/other-poultry reference-group coefficient on log distance is -0.2103 (standard error 0.0536), while the broiler interaction is $+0.2164$ (0.0568). Their sum is 0.0061, so the broiler category component almost offsets the reference-group penalty. This pattern is consistent with relatively homogeneous broiler products being easier to source over longer distances. Channel, temperature, and other distance interactions determine the remaining slope for a particular product-market observation. The positive nesting estimate, $\rho = 0.3229$ (0.0335), separately captures closer substitution within commodity groups.

With regard to supply, marginal cost is recovered jointly with demand using the multi-product pricing conditions. The baseline log-distance coefficient is 0.0280 dollars per pound, with a standard error of 0.0087 (Table A5). Because marginal cost is in levels, doubling distance changes this component by $0.0280 \log(2) = 0.0194$ dollars per pound, or about two cents. Product, shipment, channel, and year interactions modify the cost slope. The effective-distance counterfactual therefore changes both sourcing preferences and costs through the terms estimated in the two equations.

Other cost estimates connect the pricing residual to plant operations. Materials,

supplemental-input, and labor unit costs have positive coefficients; production employment and assets have negative coefficients. Frozen status has a level coefficient of 0.0821 dollars per pound and an interaction of −0.0097 with log distance, yielding a fitted frozen differential of about 0.023 dollars per pound at mean log distance 6.09. The processed-poultry level and distance terms likewise offset one another to a great extent at the mean route length. These estimates describe cost conditional on the remaining plant, product, and market characteristics.

5.4 Processor Margins and Local Procurement Demand

The common baseline has an average price of 1.443 dollars per pound and average recovered marginal cost of 0.966, leaving a price–cost wedge of 0.477 dollars per pound. The associated margin, $(p - mc)/p$, is about 33.1 percent. Recovered marginal cost is close in scale to the observed cost measure of 1.01 dollars per pound. The average own-price elasticity of a plant-product option is −3.135; holding competing prices fixed, a one-percent increase in the price of an option reduces procurement of that option by about 3.14 percent locally.

These are processors' margins on sales to retail and wholesale buyers. The model prices one layer of the chain; wholesaler and retailer margins, and any buyer power exercised downstream, are not measured. The own-price response alone does not determine the processor's margin. A buyer leaving one plant-product can choose another product belonging to the same firm. The plant-to-firm links allow the pricing conditions to account for these retained sales. Supplier differentiation, local procurement shares, and firms' sets of plant-products together determine the market power summarized by the recovered margins. The same links make it possible to study how a merger changes pricing even when the physical plants and products remain in place.

5.5 Business-Demand Orientation and Baseline Market Scale

I next relate local demand composition to the scale of procurement in normal conditions. The comparison is a post-estimation regression of model-implied county outcomes on observed county characteristics; it is not a parameter of the demand or supply model. The baseline wholesale procurement share, or PW share, is the quantity-weighted share of a county's processor-origin poultry procurement shipped to wholesalers, used here as a proxy for business-demand orientation. The comparison is a ten-percentage-point difference in that share, such as 30 rather than 20 percent. County procurement and total surplus are regressed on baseline PW share while controlling for income, education, and population. Table 4 presents these baseline comparisons alongside the corresponding loss-magnitude comparisons under the

input-cost shock.

Table 4: Business-Demand Orientation: Market Scale and Adjustment to an Input-Cost Shock

Outcome	Adjusted contrast	95% confidence interval
Panel A. Baseline equilibrium		
Procurement	+16.0%	[12.4, 19.7]%
Total surplus	+11.0%	[7.52, 15.2]%
Panel B. Raw-poultry input cost increases by 20%		
Procurement loss (signed contrast)	−6.0%	[−7.18, −4.82]%
Buyer-surplus loss (signed contrast)	−4.0%	[−4.69, −3.31]%
Additional price response	Insignificant	[−1.57, 1.98]%

Notes: Every row compares county markets with a 10-percentage-point difference in baseline PW share (for example, 30 rather than 20 percent), controlling for income, education, and population. PW share is the quantity-weighted share of processor-origin procurement shipped to wholesalers and proxies business-demand orientation. Panel A reports relative differences in baseline levels. For procurement and buyer surplus, Panel B reports relative differences in loss magnitudes under the same 20 percent input-cost shock: procurement losses are measured in pounds and buyer-surplus losses in dollars, relative to each county's model-implied baseline. The reported signs retain the direction of the underlying outcome response: −6 percent denotes a procurement loss 6 percent larger, and −4 percent a buyer-surplus loss 4 percent larger. These are differences in loss magnitudes, not percentage-point differences in decline rates. The price row concerns the additional price response associated with PW share; its interval includes zero. Intervals describe uncertainty in the county-level comparisons, conditional on model-implied equilibrium outcomes.

A ten-percentage-point higher baseline PW share is associated with 16 percent more procurement, with a 95 percent confidence interval of $[12.4, 19.7]$ percent, and 11 percent more total surplus, with an interval of $[7.52, 15.2]$ percent. Both relationships remain positive after the same local socioeconomic controls. The surplus comparison concerns the total gains from trade generated by a county's procurement, including gains to processors supplying that county.

This result links the business-serving component of procurement to market scale. Wholesalers assemble demand from food-service and institutional customers into purchases from processors, while the local mixture of sourcing decision-makers determines how that demand is allocated across plants. Counties can consequently face different procurement scales and supplier incentives even when their income, education, and population are similar. The counterfactual analysis returns to this same PW-share comparison in order to examine whether the markets with larger baseline gains also experience greater losses when input costs rise.

6 Counterfactual Simulations

The counterfactuals use the constructed network to distinguish changes in sourcing options from changes in costs and pricing incentives. Three experiments introduce shocks to the

chain by changing plant availability, effective distance, or raw-poultry input cost; the fourth experiment changes firm ownership. Each starts from the same estimated equilibrium. I first compare their effects on procurement and the allocation of surplus, then return to the baseline wholesale procurement share (PW share) to relate business-demand orientation to the response under higher input costs.

6.1 Benchmark and Welfare Aggregation

Table 5 reports annual average outcomes across the three survey years, 2007, 2012, and 2017. Baseline procurement is 21.19 billion pounds per year. Producer and downstream-buyer surplus are 11.46 and 18.45 billion dollars per year, respectively, for total gains from trade of 29.92 billion dollars. Producer surplus is assigned to the destination markets whose purchases generate it, including sales by plants in other states. The buyer component measures the value of procurement opportunities to the decision-makers represented in demand.

Table 5: Annual Average Counterfactual Outcomes

Scenario	Price	Cost	Margin (%)	Inside quantity	Producer	Buyer	Total	ΔTS	ΔTS (%)
Model-implied baseline	1.443	0.966	33.1	21.19	11.46	18.45	29.92	0.00	0.00
Plant shutdowns	1.448	0.967	33.2	20.36	11.01	17.46	28.48	-1.44	-4.81
Effective distance +20%	1.469	0.990	32.6	20.33	11.06	17.47	28.53	-1.38	-4.62
Raw-poultry cost +20%	1.558	1.085	30.4	18.09	9.09	15.53	24.62	-5.30	-17.72
Merger of third- and fourth-largest firms	1.542	0.966	37.4	19.23	12.36	16.28	28.64	-1.28	-4.27

Notes: Prices and marginal costs are in dollars per pound; inside quantity is in billion pounds per year; producer, buyer, and total surplus are in billion dollars per year. Annual averages divide the sums of the 2007, 2012, and 2017 model-implied quantities and surplus by three. Price and cost are averages across the released state rows, and margin is their price–cost wedge divided by price. ΔTS is relative to the common baseline. Entries and changes are rounded independently. Firms in the merger are ranked by CFS-expanded shipment value. All scenarios use the estimated demand and cost parameters and the common market-size normalization.
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

The baseline average price is 1.443 dollars per pound, marginal cost is 0.966, and the implied margin is 33.1 percent. State-level heterogeneity appears in Table A6 and Figure A1. Appendix F defines the quantity and surplus measures and their annual aggregation. The market-size sensitivity in Table A7 preserves the signs and ordering of the total-surplus responses across the displayed normalizations.

6.2 Counterfactual Procedure and Shock Design

As noted, three of the experiments are shocks to the chain that remove sourcing options or raise costs. The shocks are plant shutdowns, an increase in effective distance on every route, and an increase in raw-poultry cost for all processors. The fourth experiment is a merger of the third- and fourth-largest firms; it holds plants, products, and marginal costs fixed and changes only which firm prices the products. Figure 1(b) marks where each experiment acts. Estimated demand and cost parameters, fitted unobservables for available products, and market sizes are fixed in each experiment. I change the relevant primitive and solve the multiproduct pricing conditions for a new equilibrium. These are short-run responses; there is no endogenous entry or exit, and surviving plants face no capacity ceiling. The imposed shutdown removes products from the choice set; the other experiments retain the baseline offerings.

The flood-shutdown experiment removes plant-products originating in ten counties with a history of flooding. Buyers then substitute among surviving suppliers and the outside option. Appendix D.1 describes the geographic selection and public disruption examples.

The transportation experiment raises every route's effective distance by 20 percent, changing both demand and cost through all estimated distance terms and interactions. This represents detours or an equivalent increase in logistics friction, while retaining plants and destinations.

The input-cost experiment raises the raw-poultry cost component by 20 percent for all processors, adding the corresponding dollar increase to marginal cost. This first step is arithmetic rather than a model result; it raises average marginal cost by 12.3 percent. The price, quantity, and surplus changes reported below are the equilibrium responses to that higher cost. The experiment captures a common increase in raw-bird costs of the kind that followed the 2008 feed-cost surge.

The merger experiment joins the third- and fourth-largest processors under one firm identifier, holding plant-products and marginal-cost primitives fixed. One firm then prices both sets of products, accounting for purchases that switch between them. Appendix F gives the shock transformations and equilibrium calculations.

6.3 Procurement Losses and the Allocation of Surplus

The raw-poultry cost increase produces the largest decline in total gains. Average marginal cost rises by 12.3 percent, from 0.966 to 1.085 dollars per pound, while price rises by 7.9 percent to 1.558. The price–cost margin contracts from 33.1 to 30.4 percent. Procurement falls to 18.09 billion pounds per year, producer surplus falls by 20.7 percent, and buyer surplus

falls by 15.9 percent. The annual producer and buyer losses are 2.37 and 2.93 billion dollars, respectively. Total surplus falls by 5.30 billion dollars per year, or 17.72 percent. Higher input costs therefore reduce gains on both sides of the procurement market as margins and purchases contract.

The merger produces a different allocation of losses. Marginal-cost primitives remain fixed, while average price rises by 6.9 percent, to 1.542 dollars per pound, and the margin rises to 37.4 percent. Procurement falls to 19.23 billion pounds per year. Producers gain 0.90 billion dollars per year in equilibrium, while buyers lose 2.18 billion. The remaining 1.28 billion is a loss of total gains from trade; deadweight loss accounts for 59 percent of the buyer-surplus reduction. This calculation includes all modeled suppliers' equilibrium responses. It links the plant-to-firm information directly to welfare, because the same physical products now face different pricing incentives.

Plant shutdowns and greater effective distance produce smaller aggregate price responses, yet each substantially reduces total surplus. Shutdowns raise average price only to 1.448 dollars per pound, while procurement falls to 20.36 billion pounds per year and total surplus falls by 1.44 billion dollars, or 4.81 percent. Removing supplier options reduces the value of procurement even when surviving prices change little. The distance shock raises average cost to 0.990 and price to 1.469, lowers procurement to 20.33 billion pounds per year, and reduces total surplus by 1.38 billion dollars, or 4.62 percent. These similar total losses arise from different network changes; shutdowns remove selected sourcing options, while higher effective distance changes every route.

Figure 5 maps the state-level percentage changes in total surplus. Higher raw-poultry costs generate widespread losses, while the effects of shutdowns and consolidation depend more directly on local supplier exposure and overlap. Detailed state outcomes appear in Table A6; the other maps appear in Figure A2 through Figure A7.

6.4 Business-Demand Orientation and Input-Cost Exposure

The county comparison returns to the ten-percentage-point difference in baseline wholesale procurement share (PW share) used in Subsection 5.5. As there, these are post-estimation regressions controlling for income, education, and population. I relate county procurement and buyer-surplus losses under the raw-poultry cost shock to baseline PW share. Procurement loss is measured as the reduction in pounds purchased from the county's model-implied baseline, and buyer-surplus loss as the corresponding reduction in dollars. The wholesale share continues to proxy the business-serving component of procurement. The two panels of Table 4 therefore compare baseline market scale and loss magnitudes under a common

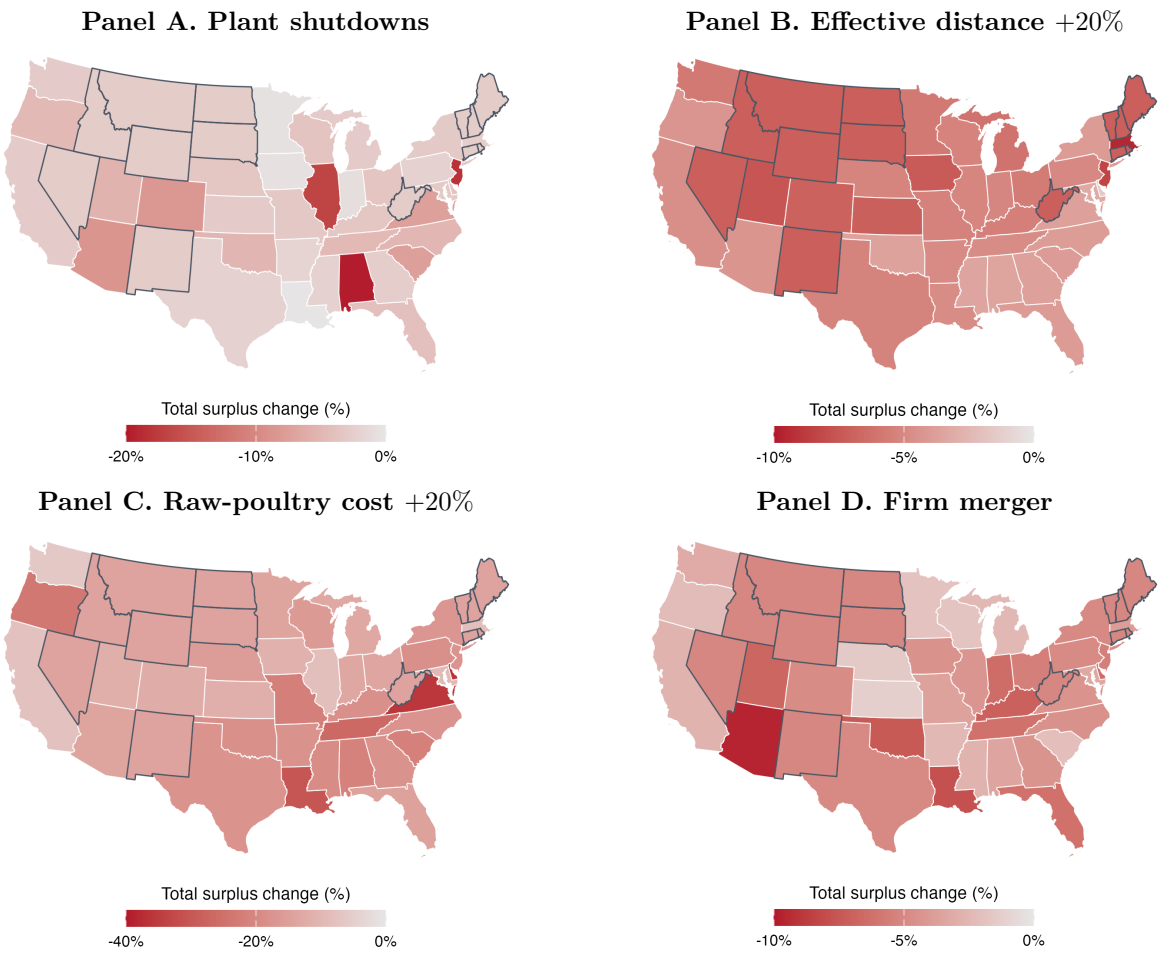


Figure 5: State-Level Total-Surplus Percentage Changes Relative to the Baseline

Notes: Total surplus equals producer surplus plus downstream-buyer surplus. Panels A–D show percentage changes relative to the model-implied baseline under plant shutdowns in the ten-county flood-history set, a 20 percent increase in effective shipment distance, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Each panel uses its own color scale with limits rounded outward to multiples of 5 percent; because every state loses surplus in all four scenarios, each panel uses a single-hue sequential scale from light gray (no change) to dark red (largest loss). Maps display the contiguous United States. States without separate state rows are colored with the "Rest of America" aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

input-cost shock using the same observable local controls.

A 10-percentage-point higher baseline PW share is associated with a procurement loss 6 percent larger in pounds and a buyer-surplus loss 4 percent larger in dollars. Table 4 retains the negative signs of the underlying outcome contrasts: -6 percent with a 95 percent confidence interval of $[-7.18, -4.82]$ percent for procurement, and -4 percent with an interval of $[-4.69, -3.31]$ percent for buyer surplus. The percentages compare loss magnitudes across county markets; they do not describe percentage-point differences in their decline rates. The interval for the additional price response is $[-1.57, 1.98]$ percent and includes zero. Against the common rise in input costs and prices, markets with a higher PW share lose more procurement volume and buyer-surplus dollars, without a statistically detectable additional price increase.

Together with the baseline results, these estimates link regional market scale to shock exposure. Counties with a larger business-oriented procurement component have greater baseline purchases and total gains from trade, and they lose more procurement and buyer surplus under higher raw-poultry costs. The model connects this pattern to the organization of sourcing decisions, local supplier alternatives, and procurement shares. Market demand combines food-service-led and wholesaler-led business procurement with retail procurement; its response follows from that mixture and the supplier network. Observing these relationships makes it possible to distinguish a cost shock that reduces trade gains from a merger that both reduces those gains and redistributes them toward processors.

7 Conclusion

This paper constructs the first shipment-based representation of the U.S. poultry procurement network to examine how supplier relationships and firm ownership shape competition and the burden of supply disruptions. The network is sparse; annual plant-to-county network densities average 10 percent for retailer-bound shipments and 14 percent for wholesaler-bound shipments. The links that exist provide essential information about local sourcing opportunities. Linked shipments identify which plants supply each destination and their procurement shares, and plant-to-firm links identify which of those alternatives are priced jointly. The wholesale channel adds an organizational distinction: processor-to-wholesaler shipments are large and long-distance, while wholesaler-origin shipments are smaller and more local.

The estimated demand-and-supply system reveals substantial processor pricing power over downstream buyers. Each processor jointly prices its products across its plants, and the baseline price–cost margin is approximately 33 percent of the selling price. Wholesale buyers

are 3.33 times as price sensitive as food-service buyers and 1.67 times as price sensitive as retailers. Limited local substitution and common ownership help account for the recovered margins, so local supplier counts alone do not describe competitive pressure. The model prices one layer of the chain: processors' margins on sales to retailers and wholesalers.

The counterfactuals distinguish the effects of reduced sourcing options and higher costs from the effects of changes in ownership. A 20 percent raw-poultry cost shock lowers both producer and buyer surplus and reduces total surplus by 17.7 percent, while a plant-shutdown scenario reduces total surplus by 4.8 percent. The simulated merger of the third- and fourth-largest processors leaves physical sourcing options and marginal costs intact but raises prices by 6.9 percent. In the case of the merger, buyers lose $2.18 billion annually, processors gain $0.90 billion, and $1.28 billion is lost from total gains from trade; deadweight loss accounts for 59 percent of the buyer loss. Input-cost inflation and consolidation thus raise prices by similar amounts but move processor surplus in opposite directions. Both reduce buyer and total surplus. Each exercise describes short-run responses with no endogenous entry or exit and no capacity constraints; the outcomes trace back to which procurement alternatives remain, what they cost, and which firms price them.

The burden falls unevenly across markets. In post-estimation comparisons controlling for income, education, and population, a 10-percentage-point higher baseline wholesale procurement share is associated with 16 percent more procurement and 11 percent more total surplus. Under the common 20 percent raw-poultry cost increase, a 10-percentage-point higher baseline wholesale procurement share is associated with a procurement loss 6 percent larger in pounds and a buyer-surplus loss 4 percent larger in dollars, without a statistically detectable additional price response. These are relative differences in magnitudes of loss. Business-oriented procurement markets are larger at baseline and lose more purchases and surplus dollars under a common cost increase, making prices an incomplete measure of their burden.

The empirical contribution is a path from observed shipments to local choice sets, from choice sets and ownership to pricing incentives, and from buyer composition to the regional incidence of shocks. Measuring that intermediate procurement layer makes it possible to ask competition and welfare questions that endpoint geography cannot answer. The resulting evidence connects wholesale demand organization, supplier geography, and ownership to regional procurement scale and the allocation of shock losses in the poultry industry.

Data Availability and Disclosure Statement

The empirical analysis relies on restricted-use Census microdata and linked establishment identifiers. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2521 and was reviewed for release under Disclosure Clearance Numbers CBDRB-FY25-0219 and CBDRB-FY26-0215. Public documentation for the CFS and Census business surveys is available from the U.S. Census Bureau and the Bureau of Transportation Statistics. The restricted microdata cannot be redistributed by the author. The replication package will contain code, synthetic or public-use examples where feasible, and disclosure-cleared tables and figures. All tables, figures, and other reported statistics in the public manuscript are subject to Census Bureau disclosure review and should be cited with the project and disclosure clearance numbers reported in this manuscript.

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A Tables and Figures

A.1 Tables

Table A1: Full-Sample Fitted Confusion Matrix for Final PR–PW Classifier

	Reference label	
Prediction	PW label	PR label
PW label	∼1,200	∼80
PR label	∼100	∼15,500

Notes: Entries are rounded, disclosure-cleared fitted counts from the final random forest after refitting on the full mechanically labeled sample. This is not a held-out confusion matrix. Subsection 3.2 describes the 90–10 training-validation procedure used to select the specification and the subsequent full-sample refit. Accuracy on the 10 percent validation set used during specification selection was about 96 percent. Class frequencies are record counts in the mechanically labeled sample, not national physical-flow shares.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

Table A2: Descriptive Statistics for Estimation Variables

Variable	Mean	Std. dev.
price	1.4230	0.177
Share	0.0458	0.073
Log distance	6.0900	1.113
Frozen-shipment indicator	0.2769	0.451
Broiler indicator	0.6000	0.492
Processed-poultry indicator	0.2462	0.427
Private-carriage share (private)	0.1739	0.363
Wholesale-channel share (ToW)	0.0931	0.282
Log total production workers	7.9740	0.669
Log asset stock	10.7500	1.161
Materials unit cost ($/lb)	0.6860	0.395
Supplemental unit cost ($/lb)	0.1677	0.153
Labor unit cost ($/lb)	0.1500	0.097
Capital expenditure rate	0.2434	0.242
Production-worker hourly pay	13.5000	3.136
Inventory-to-assets ratio	0.0309	0.040
Own-slaughter plant	0.6000	0.492
External-slaughter plant	0.1308	0.339
Rival product count	11.8500	10.521
Rival frozen-product count	4.7080	6.990
Rival broiler count	6.1320	6.023
Rival processed-product count	4.2940	7.861

Notes: The sample contains 6,500 product-market observations, 700 markets, and 400 products. Reported statistics are rounded in accordance with Census Bureau disclosure review, so distinct variables can display identical rounded values. The price and unit-cost variables are in constant 2017 dollars per pound; production-worker hourly pay is in constant 2017 dollars per hour. Means are simple across product-market observations; in particular, mean ToW is not a nationally aggregated PW weight share.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

Table A3: Sampled Shipments per Product-Market Cell

Shipments per cell	Share of cells	Mean shipments	Share of CFS-expanded weight
1–2	29.5%	1.5	4.5%
3–4	25.4%	3.6	12.4%
5–9	23.7%	6.7	26.2%
10–19	12.9%	13.0	24.6%
20 or more	8.5%	23.7	32.3%

Notes: The table reports the distribution of sampled CFS shipment records underlying the 6,500 product-market cells in the estimation sample. Across cells, the mean is about 6.6 shipments, the median is 4, the 90th and 95th percentiles are 15 and 22, and the maximum is about 55; the weight-weighted median cell is built from about 12 shipments. Shares and counts are rounded in accordance with Census Bureau disclosure review.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

Table A4: BLP Mean Utility and Nested-Logit Parameters with Fixed Effects

Variable	Estimate
Frozen-shipment indicator	0.3272
	(0.2842)
Broiler product indicator	-0.9779***
	(0.3638)
Processed-poultry indicator	-0.3449
	(0.3992)
private	-0.1787
	(0.2856)
Retail share (ToR)	-1.4330**
	(0.4908)
Log shipment distance	-0.2103***
	(0.0536)
Frozen × log distance	-0.0420
	(0.0446)
Broiler × log distance	0.2164***
	(0.0568)
Processed poultry × log distance	0.0919
	(0.0628)
private × log distance	0.0345
	(0.0500)
Retail share × log distance	-0.1433*
	(0.0766)
Nested-logit parameter (ρ)	0.3229***
	(0.0335)
Firm fixed effects	Yes
Origin-state-by-survey-year fixed effects	Yes
Destination-state-by-survey-year fixed effects	Yes

Notes: The estimation sample contains 6,500 product-market observations in 700 destination-county-year markets with 400 products. Standard errors clustered by destination county are in parentheses. The BLP demand specification absorbs LBD firm, producing-establishment origin-state-by-survey-year, and destination-state-by-survey-year fixed effects; the intercept and standalone year indicators are therefore not separately reported. These effects apply to demand, not to the supply equation. The BLP nesting parameter ρ is reported with the demand-side parameters because it governs within-nest substitution rather than marginal cost. It uses the same broiler, processed-poultry, and turkey/other-poultry nests as the preliminary nested-logit model. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

Table A5: Supply-Side Cost Parameters

Variable	Estimate	Variable	Estimate
Constant	0.6170***	Log shipment distance	0.0280***
	(0.1060)		(0.0087)
Log total production workers	-0.0146**	Frozen × log distance	-0.0097
	(0.0061)		(0.0059)
Log asset stock	-0.0111***	Broiler × log distance	-0.0027
	(0.0042)		(0.0079)
Materials unit cost ($/lb)	0.0885***	Processed poultry × log distance	0.0238***
	(0.0083)		(0.0090)
Supplemental unit cost ($/lb)	0.1902***	private × log distance	-0.0078
	(0.0202)		(0.0066)
Labor unit cost ($/lb)	0.1893***	ToW × log distance	0.0082
	(0.0331)		(0.0087)
Capital expenditure rate	0.0693***	2012 indicator	0.0087
	(0.0167)		(0.0456)
Production-worker hourly pay	0.0003	2017 indicator	0.2036***
	(0.0013)		(0.0385)
Inventory-to-assets ratio	0.5900***	2012 indicator × log distance	0.0047
	(0.1002)		(0.0073)
Own-slaughter plant	-0.0122*	2017 indicator × log distance	-0.0239***
	(0.0071)		(0.0059)
External-slaughter plant	0.1076***	Log destination-county population	0.0195***
	(0.0102)		(0.0034)
Frozen-shipment indicator	0.0821**	Rival product count	0.0015*
	(0.0375)		(0.0009)
Broiler product indicator	-0.0509	Rival frozen-product count	0.0024**
	(0.0494)		(0.0010)
Processed-poultry indicator	-0.1453**	Rival broiler count	0.0023*
	(0.0567)		(0.0013)
private	0.0442	Rival processed-product count	-0.0026**
	(0.0377)		(0.0011)
ToW	-0.0297		
	(0.0552)		

Notes: The estimation sample contains 6,500 product-market observations in 700 destination-county-year markets with 400 products. Standard errors clustered by destination county are in parentheses. The dependent variable is marginal cost in levels recovered from the supply-side first-order conditions; the cost parameters are estimated jointly with the demand parameters in the GMM system. The firm, origin-state-by-survey-year, and destination-state-by-survey-year fixed effects used in demand are not imposed on this supply equation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

Table A6: State-Level Counterfactual Results by Scenario: Panel A. Baseline
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

State	Price	Cost	Elas.	Mkt.	Inside	PS	BS	TS
Alabama	1.426	0.9439	-3.063	2.155	0.9353	0.5223	0.8417	1.364
Arizona	1.429	0.9961	-3.443	0.5847	0.3021	0.1402	0.2577	0.398
Arkansas	1.416	0.9202	-3.377	7.083	1.936	1.168	1.631	2.799
California	1.436	0.9876	-3.199	4.15	1.936	0.974	1.706	2.68
Colorado	1.472	1.032	-3.33	0.3663	0.1575	0.0733	0.1356	0.2089
Delaware	1.378	0.8349	-3.03	0.6247	0.3337	0.2233	0.3057	0.529
Florida	1.444	0.9838	-3.388	2.641	1.279	0.659	1.119	1.778
Georgia	1.413	0.9164	-3.195	5.537	2.14	1.097	1.801	2.898
Illinois	1.5	1.042	-3.378	1.712	0.8507	0.3987	0.77	1.169
Indiana	1.436	0.9509	-3.073	0.2931	0.1308	0.07023	0.1177	0.1879
Iowa	1.468	0.9756	-3.148	0.2647	0.1439	0.08323	0.1394	0.2226
Kansas	1.451	0.9696	-3.283	0.389	0.1518	0.0834	0.1354	0.2188
Kentucky	1.504	1.01	-3.088	0.3813	0.2078	0.1067	0.1847	0.2914
Louisiana	1.379	0.8809	-2.842	0.6057	0.2781	0.164	0.2402	0.4043
Maryland	1.418	0.9756	-3.244	0.7873	0.4493	0.2189	0.419	0.638
Massachusetts	1.495	1.079	-3.275	0.2173	0.1215	0.05487	0.1127	0.1676
Michigan	1.489	1.024	-3.194	0.9067	0.4153	0.2051	0.3797	0.5847
Minnesota	1.398	0.9422	-3.116	0.6477	0.258	0.1164	0.2105	0.327
Mississippi	1.415	0.9023	-3.055	1.545	0.6197	0.3287	0.535	0.8637
Missouri	1.421	0.917	-3.071	0.6993	0.3443	0.2109	0.3095	0.5203
Nebraska	1.477	0.9665	-3.321	0.1125	0.0609	0.0406	0.05753	0.09813
New Jersey	1.429	0.9899	-3.081	0.5577	0.2599	0.1236	0.2198	0.3433
New York	1.422	0.9719	-3.142	2.229	1.009	0.4627	0.8547	1.317
North Carolina	1.408	0.9215	-2.99	1.471	0.7557	0.4007	0.6663	1.067
Ohio	1.446	0.977	-3.061	0.8693	0.4453	0.2328	0.402	0.635
Oklahoma	1.45	0.9356	-3.065	0.2054	0.09523	0.057	0.08373	0.1407
Oregon	1.469	1.014	-2.753	0.2669	0.1428	0.08737	0.1252	0.2126
Pennsylvania	1.433	0.9741	-3.074	2.013	0.8163	0.4053	0.6903	1.096
South Carolina	1.442	0.9649	-2.863	0.4737	0.2376	0.1275	0.2084	0.336
Tennessee	1.451	0.9185	-2.932	0.7487	0.359	0.2167	0.3102	0.527
Texas	1.397	0.9138	-3.318	4.933	2.006	1.075	1.705	2.78
Utah	1.506	1.062	-3.257	0.3232	0.1718	0.0797	0.1494	0.2291
Virginia	1.418	0.9137	-2.535	3.477	1.173	0.9037	1.033	1.937
Washington	1.478	1.02	-3.432	0.374	0.2048	0.09383	0.1829	0.2767
Wisconsin	1.478	0.9795	-2.966	0.516	0.2688	0.1528	0.2441	0.397
Rest of America	1.472	0.9683	-3.287	0.3897	0.1893	0.1052	0.1689	0.2741

Table A6: State-Level Counterfactual Results by Scenario: Panel B. Plant shutdowns
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

State	Price	Cost	Elas.	Mkt.	Inside	PS	BS	TS
Alabama	1.412	0.9364	-2.985	2.155	0.8237	0.4247	0.6693	1.094
Arizona	1.429	0.9923	-3.368	0.5847	0.2833	0.1322	0.2312	0.3633
Arkansas	1.419	0.9181	-3.365	7.083	1.889	1.152	1.593	2.745
California	1.443	0.9903	-3.183	4.15	1.877	0.9543	1.65	2.604
Colorado	1.493	1.045	-3.325	0.3663	0.1423	0.06733	0.1239	0.1913
Delaware	1.398	0.8573	-2.901	0.6247	0.3265	0.2164	0.2932	0.5097
Florida	1.453	0.9885	-3.375	2.641	1.232	0.642	1.064	1.706
Georgia	1.417	0.9156	-3.164	5.537	2.084	1.083	1.738	2.821
Illinois	1.511	1.058	-3.572	1.712	0.7493	0.314	0.657	0.971
Indiana	1.442	0.9542	-3.075	0.2931	0.1291	0.06957	0.1166	0.1862
Iowa	1.476	0.9871	-3.107	0.2647	0.1435	0.08253	0.1388	0.2213
Kansas	1.457	0.9708	-3.268	0.389	0.1471	0.0823	0.1302	0.2125
Kentucky	1.501	1.011	-3.047	0.3813	0.2023	0.103	0.1785	0.2816
Louisiana	1.372	0.8733	-2.839	0.6057	0.2776	0.1638	0.2396	0.4033
Maryland	1.427	0.9795	-3.191	0.7873	0.439	0.2153	0.4033	0.6187
Massachusetts	1.48	1.061	-3.201	0.2173	0.1195	0.0543	0.109	0.1633
Michigan	1.486	1.02	-3.178	0.9067	0.4057	0.1991	0.3683	0.5673
Minnesota	1.397	0.9403	-3.107	0.6477	0.2568	0.1164	0.209	0.3254
Mississippi	1.408	0.8914	-3.017	1.545	0.605	0.327	0.5173	0.8443
Missouri	1.424	0.9117	-3.073	0.6993	0.334	0.2072	0.2968	0.504
Nebraska	1.477	0.94	-3.324	0.1125	0.05777	0.03973	0.05503	0.09477
New Jersey	1.434	0.9956	-3.323	0.5577	0.2277	0.1012	0.1792	0.2804
New York	1.427	0.9748	-3.127	2.229	0.9813	0.4543	0.8233	1.278
North Carolina	1.424	0.9299	-2.956	1.471	0.7187	0.391	0.6223	1.013
Ohio	1.449	0.9773	-3.006	0.8693	0.4267	0.2319	0.3803	0.6123
Oklahoma	1.456	0.937	-3.001	0.2054	0.09063	0.05577	0.07773	0.1335
Oregon	1.492	1.025	-2.7	0.2669	0.1358	0.08607	0.1162	0.2023
Pennsylvania	1.438	0.9776	-3.043	2.013	0.7983	0.4003	0.6723	1.073
South Carolina	1.469	0.9798	-2.844	0.4737	0.2213	0.121	0.1893	0.3103
Tennessee	1.429	0.8916	-2.835	0.7487	0.343	0.2137	0.2881	0.502
Texas	1.392	0.9102	-3.295	4.933	1.968	1.06	1.658	2.718
Utah	1.507	1.062	-3.223	0.3232	0.1644	0.0768	0.1397	0.2165
Virginia	1.422	0.9235	-2.597	3.477	1.119	0.829	0.9617	1.791
Washington	1.487	1.028	-3.412	0.374	0.1995	0.09253	0.1763	0.2688
Wisconsin	1.488	0.978	-2.946	0.516	0.2596	0.151	0.2315	0.3823
Rest of America	1.494	0.9668	-3.288	0.3897	0.1822	0.1037	0.1627	0.2665

Table A6: State-Level Counterfactual Results by Scenario: Panel C. Effective distance +20%
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

State	Price	Cost	Elas.	Mkt.	Inside	PS	BS	TS
Alabama	1.449	0.966	-3.086	2.155	0.9077	0.507	0.8093	1.316
Arizona	1.467	1.03	-3.459	0.5847	0.29	0.1354	0.2452	0.3807
Arkansas	1.439	0.9431	-3.398	7.083	1.85	1.116	1.545	2.662
California	1.469	1.017	-3.214	4.15	1.847	0.9503	1.605	2.556
Colorado	1.507	1.064	-3.358	0.3663	0.148	0.06923	0.125	0.1943
Delaware	1.395	0.8505	-3.049	0.6247	0.3286	0.221	0.2996	0.5207
Florida	1.474	1.011	-3.411	2.641	1.229	0.6403	1.066	1.706
Georgia	1.436	0.9384	-3.217	5.537	2.075	1.064	1.727	2.79
Illinois	1.528	1.066	-3.397	1.712	0.812	0.3843	0.724	1.108
Indiana	1.464	0.9756	-3.099	0.2931	0.1246	0.06783	0.1105	0.1783
Iowa	1.49	0.997	-3.194	0.2647	0.1365	0.07823	0.128	0.2063
Kansas	1.483	0.9996	-3.33	0.389	0.1424	0.0786	0.1246	0.2032
Kentucky	1.53	1.037	-3.119	0.3813	0.1988	0.102	0.1733	0.2753
Louisiana	1.403	0.9029	-2.881	0.6057	0.2676	0.157	0.2278	0.385
Maryland	1.446	1	-3.281	0.7873	0.4363	0.2143	0.403	0.6173
Massachusetts	1.517	1.096	-3.247	0.2173	0.1126	0.05133	0.1001	0.1515
Michigan	1.52	1.051	-3.24	0.9067	0.396	0.1954	0.3533	0.5487
Minnesota	1.415	0.958	-3.157	0.6477	0.2452	0.1113	0.1967	0.308
Mississippi	1.432	0.9195	-3.073	1.545	0.6007	0.3193	0.514	0.8333
Missouri	1.448	0.9419	-3.097	0.6993	0.3295	0.2014	0.291	0.4923
Nebraska	1.503	0.9885	-3.337	0.1125	0.0584	0.03903	0.05397	0.093
New Jersey	1.442	1.001	-3.083	0.5577	0.2425	0.1151	0.1994	0.3146
New York	1.45	0.9966	-3.171	2.229	0.974	0.4483	0.816	1.264
North Carolina	1.428	0.9405	-3.007	1.471	0.7327	0.3893	0.636	1.025
Ohio	1.472	1.001	-3.106	0.8693	0.425	0.2227	0.3757	0.5983
Oklahoma	1.477	0.9607	-3.066	0.2054	0.0919	0.05557	0.08003	0.1356
Oregon	1.499	1.041	-2.789	0.2669	0.1371	0.08507	0.1183	0.2034
Pennsylvania	1.458	0.9967	-3.085	2.013	0.782	0.3883	0.652	1.04
South Carolina	1.461	0.9832	-2.892	0.4737	0.2305	0.1229	0.1995	0.3224
Tennessee	1.479	0.9468	-2.952	0.7487	0.3457	0.2071	0.2943	0.5013
Texas	1.427	0.9414	-3.349	4.933	1.901	1.036	1.596	2.633
Utah	1.541	1.092	-3.264	0.3232	0.1605	0.0757	0.1359	0.2116
Virginia	1.443	0.9371	-2.541	3.477	1.138	0.8737	0.991	1.865
Washington	1.509	1.045	-3.476	0.374	0.1941	0.09067	0.17	0.2606
Wisconsin	1.5	1.001	-2.986	0.516	0.2569	0.1463	0.2297	0.376
Rest of America	1.49	0.9857	-3.289	0.3897	0.1784	0.1001	0.1543	0.2545

Table A6: State-Level Counterfactual Results by Scenario: Panel D. Raw-poultry cost +20%
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

State	Price	Cost	Elas.	Mkt.	Inside	PS	BS	TS
Alabama	1.566	1.088	-3.522	2.155	0.7707	0.3947	0.6763	1.071
Arizona	1.524	1.087	-3.795	0.5847	0.2659	0.1182	0.2232	0.3413
Arkansas	1.556	1.054	-3.623	7.083	1.639	0.9347	1.362	2.297
California	1.507	1.058	-3.353	4.15	1.784	0.87	1.601	2.471
Colorado	1.546	1.106	-3.668	0.3663	0.1409	0.06133	0.1219	0.1832
Delaware	1.654	1.132	-3.773	0.6247	0.2365	0.1367	0.1988	0.3353
Florida	1.548	1.091	-3.68	2.641	1.134	0.5353	0.9823	1.518
Georgia	1.566	1.071	-3.564	5.537	1.801	0.8793	1.496	2.375
Illinois	1.521	1.063	-3.6	1.712	0.801	0.3633	0.7087	1.072
Indiana	1.556	1.065	-3.414	0.2931	0.1144	0.06007	0.1027	0.1627
Iowa	1.553	1.081	-3.417	0.2647	0.1331	0.0677	0.1302	0.1979
Kansas	1.594	1.101	-3.507	0.389	0.1316	0.07393	0.1192	0.1932
Kentucky	1.562	1.077	-3.508	0.3813	0.1762	0.08803	0.1499	0.238
Louisiana	1.563	1.063	-3.522	0.6057	0.2069	0.1122	0.1687	0.281
Maryland	1.533	1.086	-3.416	0.7873	0.406	0.2009	0.3763	0.5773
Massachusetts	1.507	1.094	-3.583	0.2173	0.1145	0.04927	0.1044	0.1537
Michigan	1.547	1.091	-3.562	0.9067	0.3743	0.1696	0.3373	0.507
Minnesota	1.548	1.086	-3.468	0.6477	0.2222	0.1003	0.1801	0.2805
Mississippi	1.604	1.092	-3.545	1.545	0.5033	0.2663	0.4307	0.697
Missouri	1.577	1.082	-3.456	0.6993	0.2815	0.1635	0.2427	0.4063
Nebraska	1.563	1.076	-3.605	0.1125	0.05387	0.03337	0.04947	0.08283
New Jersey	1.544	1.099	-3.576	0.5577	0.2206	0.09697	0.1858	0.2828
New York	1.544	1.091	-3.511	2.229	0.854	0.3793	0.7137	1.093
North Carolina	1.551	1.075	-3.408	1.471	0.641	0.3155	0.563	0.8783
Ohio	1.535	1.073	-3.42	0.8693	0.396	0.1969	0.3517	0.5487
Oklahoma	1.542	1.05	-3.447	0.2054	0.08503	0.0443	0.07057	0.1149
Oregon	1.555	1.102	-3.111	0.2669	0.118	0.06313	0.0995	0.1626
Pennsylvania	1.555	1.091	-3.464	2.013	0.6863	0.328	0.5653	0.8933
South Carolina	1.597	1.12	-3.379	0.4737	0.1959	0.09633	0.1666	0.2629
Tennessee	1.59	1.086	-3.557	0.7487	0.2841	0.1502	0.2368	0.387
Texas	1.567	1.083	-3.648	4.933	1.692	0.8477	1.437	2.285
Utah	1.538	1.1	-3.544	0.3232	0.1554	0.07073	0.1312	0.202
Virginia	1.574	1.077	-3.43	3.477	0.8717	0.5273	0.7127	1.24
Washington	1.572	1.112	-3.65	0.374	0.1921	0.08723	0.1716	0.2589
Wisconsin	1.556	1.076	-3.38	0.516	0.2399	0.122	0.2108	0.3328
Rest of America	1.567	1.075	-3.446	0.3897	0.1687	0.08787	0.1469	0.2347

Table A6: State-Level Counterfactual Results by Scenario: Panel E. Firm merger
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

State	Price	Cost	Elas.	Mkt.	Inside	PS	BS	TS
Alabama	1.512	0.9439	-3.14	2.155	0.8637	0.562	0.7547	1.317
Arizona	1.583	0.9961	-3.799	0.5847	0.2435	0.1636	0.1958	0.3593
Arkansas	1.452	0.9202	-3.463	7.083	1.831	1.214	1.516	2.73
California	1.506	0.9876	-3.28	4.15	1.812	1.035	1.573	2.608
Colorado	1.546	1.032	-3.505	0.3663	0.142	0.08133	0.1175	0.1989
Delaware	1.567	0.8349	-3.248	0.6247	0.2873	0.2447	0.2499	0.4947
Florida	1.568	0.9838	-3.602	2.641	1.113	0.7337	0.9333	1.667
Georgia	1.493	0.9164	-3.308	5.537	1.943	1.193	1.578	2.771
Illinois	1.569	1.042	-3.525	1.712	0.7707	0.4447	0.6737	1.118
Indiana	1.568	0.9509	-3.28	0.2931	0.1136	0.0783	0.0974	0.1757
Iowa	1.58	0.9756	-3.265	0.2647	0.13	0.0907	0.122	0.2127
Kansas	1.49	0.9696	-3.326	0.389	0.1469	0.085	0.1312	0.2162
Kentucky	1.675	1.01	-3.283	0.3813	0.1767	0.12	0.1508	0.2708
Louisiana	1.497	0.8809	-3.097	0.6057	0.2332	0.1822	0.1903	0.3727
Maryland	1.514	0.9756	-3.355	0.7873	0.4163	0.2384	0.3783	0.6167
Massachusetts	1.628	1.079	-3.432	0.2173	0.1108	0.0611	0.09993	0.161
Michigan	1.538	1.024	-3.283	0.9067	0.3927	0.2179	0.3527	0.5707
Minnesota	1.419	0.9422	-3.19	0.6477	0.2478	0.1223	0.1984	0.3207
Mississippi	1.482	0.9023	-3.142	1.545	0.582	0.3453	0.495	0.8403
Missouri	1.495	0.917	-3.165	0.6993	0.3162	0.2242	0.2773	0.5013
Nebraska	1.569	0.9665	-3.364	0.1125	0.0588	0.04137	0.0553	0.09667
New Jersey	1.5	0.9899	-3.304	0.5577	0.2312	0.1364	0.1881	0.3245
New York	1.548	0.9719	-3.3	2.229	0.9063	0.5097	0.7423	1.252
North Carolina	1.523	0.9215	-3.108	1.471	0.6813	0.4347	0.584	1.019
Ohio	1.579	0.977	-3.227	0.8693	0.395	0.2569	0.343	0.6
Oklahoma	1.625	0.9356	-3.302	0.2054	0.0809	0.06277	0.06753	0.1303
Oregon	1.544	1.014	-2.811	0.2669	0.1348	0.09077	0.1171	0.2078
Pennsylvania	1.516	0.9741	-3.219	2.013	0.734	0.444	0.598	1.042
South Carolina	1.521	0.9649	-2.898	0.4737	0.2262	0.1323	0.1963	0.3286
Tennessee	1.615	0.9185	-3.067	0.7487	0.3108	0.2358	0.2582	0.494
Texas	1.492	0.9138	-3.484	4.933	1.781	1.162	1.48	2.642
Utah	1.637	1.062	-3.556	0.3232	0.1479	0.09077	0.1227	0.2135
Virginia	1.514	0.9137	-2.545	3.477	1.058	0.956	0.8943	1.85
Washington	1.56	1.02	-3.561	0.374	0.19	0.1007	0.1673	0.268
Wisconsin	1.53	0.9795	-3.02	0.516	0.2569	0.1581	0.2315	0.3897
Rest of America	1.571	0.9683	-3.486	0.3897	0.1677	0.1149	0.1453	0.2602

Table A7: Sensitivity to the Market-Size Normalization

								ΔTS (%) by scenario			
κ	Mkt.	Inside share	Elas.	Margin (%)	PS	BS	TS	Plant	Distance	Raw cost	Merger
0.6	58.98	0.359	-3.180	32.8	11.37	18.25	29.62	-4.57	-4.54	-17.13	-4.02
0.7 (baseline)	50.55	0.419	-3.135	33.1	11.46	18.45	29.92	-4.81	-4.62	-17.72	-4.27
0.8	44.23	0.479	-3.041	33.7	11.58	19.41	30.99	-5.02	-4.71	-18.17	-4.41

Notes: Each row re-estimates the full model and re-solves all four counterfactual scenarios with the outside-option normalization in Appendix B.5 set to κ in place of the baseline 0.7. Mkt. is mean annual aggregate market size in billion lb and Inside share is aggregate inside sales divided by market size; both follow mechanically from the normalization. Elas. is the average own-price elasticity; Margin is the average price–cost difference as a percent of the price. PS, BS, and TS are baseline producer, downstream-buyer, and total surplus in billions of dollars per year under each normalization. Annual aggregates average the 2007, 2012, and 2017 model outcomes. The ΔTS columns report each scenario's total-surplus change relative to that row's baseline, in percent. Scenario rankings and signs are unchanged across normalizations.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

Table A8: Public Examples of Weather-Disrupted Poultry-Processing and Production Counties

County	FIPS	Poultry-processing anchor	Weather or disruption motivation
Blount County, AL	01009	Tyson Blountsville chicken-processing plant.	April 2011 tornado outbreak: Tyson reported that the Blountsville plant was idled after losing power.
Marshall County, AL	01095	Tyson Albertville chicken-processing plant and Wayne Farms Albertville processing facilities.	April 2011 tornado outbreak: Tyson reported that Albertville was idled after losing power; Wayne Farms reported Albertville-area processing facilities idled for seven days.
Morgan County, AL	01103	Wayne Farms Decatur processing facilities.	April 2011 tornado outbreak: Wayne Farms reported that Decatur-area processing facilities were idled for seven days by widespread power outages.
Tuscaloosa County, AL	01125	Peco Foods Tuscaloosa poultry facility.	April 2011 tornado outbreak: Meat+Poultry reported that the Tuscaloosa facility was heavily damaged.
Forsyth County, GA	13117	Tyson Cumming chicken plant.	Hurricane Irma in September 2017: Tyson shut down the Cumming chicken plant in response to the storm.
Terrell County, GA	13273	Tyson Dawson chicken plant.	Hurricane Irma in September 2017: Tyson shut down the Dawson chicken plant in response to the storm.
Dooly County, GA	13093	Tyson Vienna chicken plant.	Hurricane Irma in September 2017: Tyson shut down the Vienna chicken plant in response to the storm.
Lenoir County, NC	37107	Sanderson Farms Kinston poultry-processing facility.	Hurricane Matthew in October 2016: Reuters reported severe North Carolina poultry losses, including about 250,000 Sanderson broilers lost in the state.
Lexington County, SC	45063	House of Raeford West Columbia poultry plant and South Carolina grower region.	Hurricane Matthew in October 2016: House of Raeford reported partial contract-farm flock losses and weather-related road-access problems.
Jasper County, MS	28061	Peco Foods Bay Springs poultry-processing and slaughter facility.	April 2011 tornado outbreak: the National Weather Service reported significant poultry-farm damage along the Smith–Jasper–Clarke tornado track.

Notes: The table reports public examples of southeastern counties that combine poultry-processing capacity or upstream poultry production with documented weather-related disruptions, including floods, storms, and tornadoes. It does not disclose the confidential set of ten flood-history counties used in the plant-shutdown simulation. The examples illustrate the type of geography used to motivate the stylized choice-set shock: counties where storms, flooding, power outages, road-access problems, or live-bird losses plausibly interrupted poultry production or processing. Sources: Tyson Foods (2011), MEAT+POULTRY (2011), Powder & Bulk Solids (2017), AgCanada/Reuters (2016), WATTAgNet (2016), National Weather Service Jackson (2011), FSIS (n.d.a), FSIS (n.d.b), and FSIS (n.d.c).
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

Table A9: Main Variables Used in Estimation
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

Variable	Definition
price	CFS-expanded shipment value divided by expanded shipment weight, using the within-establishment annualization weight and aggregated to the product-market-year level in constant 2017 dollars per pound. CFS shipment value excludes freight charges and excise taxes.
Share	Product-market quantity divided by adjusted market size.
Log distance	Log of shipment distance weighted by CFS-expanded shipment weight.
Frozen	Indicator equal to one for SCTG 05122 and zero for SCTG 05121.
Broiler	Indicator for plants assigned to the broiler product category using manufacturing product-line information.
Processed poultry	Indicator for plants assigned to the processed-poultry product category; the omitted category is the turkey/other-poultry group.
private	Private-carriage share: the share of shipments transported using shipper-owned trucks.
ToW	Product-market CFS-expanded weight share of processor-origin shipments classified as processor-to-wholesaler (PW) using the PR–PW classifier.
ToR	Product-market retail-channel share, equal to $1 - ToW_{jm}$; enters demand in levels and interacted with log distance.
$PWShare_m^{\text{base}}$	Baseline county PW pounds divided by total processor-origin procurement; the business-demand-orientation measure in Table 4.
$\Pr(t \mid m)$	Market-specific probability of a procurement decision by type t, proportional to local LBD employment times the adjusted type factor.
type_R	Sampled-record indicator equal to one for retail representative buyer records; food service is the omitted buyer type.
type_W	Sampled-record indicator equal to one for wholesale representative buyer records; food service is the omitted buyer type.
sampling_weight	LBD procurement-decision sampling weight: establishment employment times its adjusted type factor (0.6 for food service, 2.9 for retail, 15.2 for wholesale); normalize within the 100-mile market area.
tpw	Total production workers.
tae	Plant asset stock measure.
wp	Production-worker payroll per production-worker hour.
te	Capital expenditures divided by plant assets.
mc	Materials unit cost, equal to materials cost divided by CFS-expanded shipment weight and reported in $/lb.
sc	Supplemental input unit cost, equal to supplemental input cost divided by CFS-expanded shipment weight and reported in $/lb.
wc	Labor unit cost, equal to labor cost divided by CFS-expanded shipment weight and reported in $/lb.
tie	Total inventories divided by ending total assets.
Packaging input intensity	CMF materials-detail packaging cost divided by CFS-expanded shipment weight ($/lb); enters the excluded demand instruments.
Raw-poultry input intensity	CMF materials-detail raw-poultry cost divided by CFS-expanded shipment weight ($/lb); enters the excluded demand instruments and is the cost term raised by 20 percent in the raw-poultry counterfactual.
Poultry-parts input intensity	CMF materials-detail processed-poultry cost divided by CFS-expanded shipment weight ($/lb); enters the excluded demand instruments.
TYPOP104	Indicator for poultry or meat products processed chiefly from animals slaughtered in the same establishment.
TYPOP106	Indicator for poultry or meat products processed chiefly from animals slaughtered elsewhere.
Turkey indicator	Indicator equal to one for plants whose CMF product line is turkey; used only in constructing the local differentiation instruments (turkey plants belong to the other-poultry category).

Continued on next page

Table A9: Main Variables Used in Estimation (continued)
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

Variable	Definition
Log destination-county population	Log of population in the market's destination county and year; a supply-side control in the marginal-cost equation.
n_rival	Number of competing products in the same destination-county-year market.
n_rival_frozen	Number of competing frozen products in the same market.
n_rival_broiler	Number of competing broiler products in the same market.
n_rival_processed	Number of competing processed-poultry products in the same market.

A.2 Figures

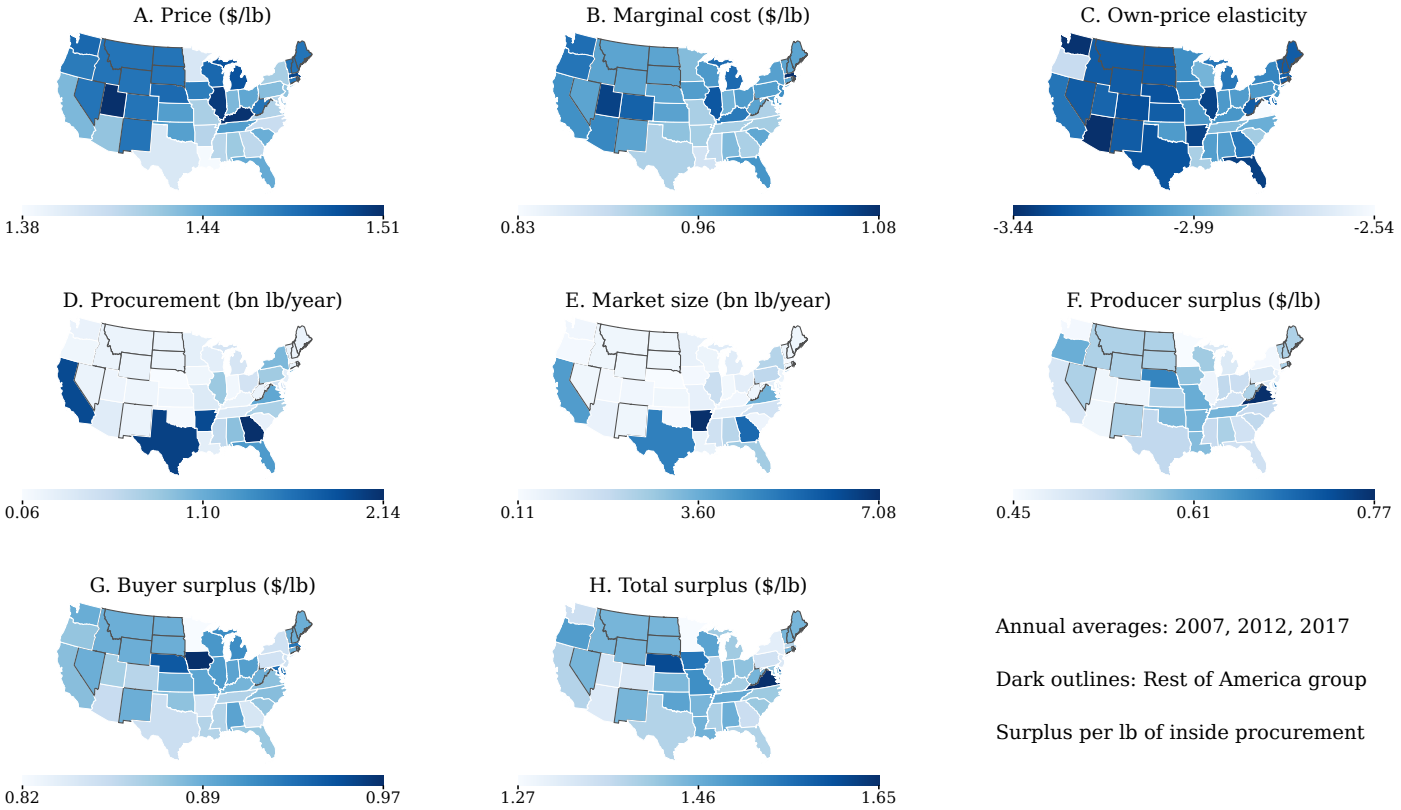


Figure A1: Model-Implied Baseline State-Level Distributions

Notes: Model-implied state outcomes. Price, marginal cost, and surplus per pound are in $/lb; procurement and market size are annual averages in billion lb across 2007, 2012, and 2017. Total surplus is producer plus buyer surplus. Each panel has its own continuous color scale. Darker shades indicate larger values, except for elasticity, where they indicate more elastic demand. States without separate rows take the Rest of America group value and have dark outlines. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

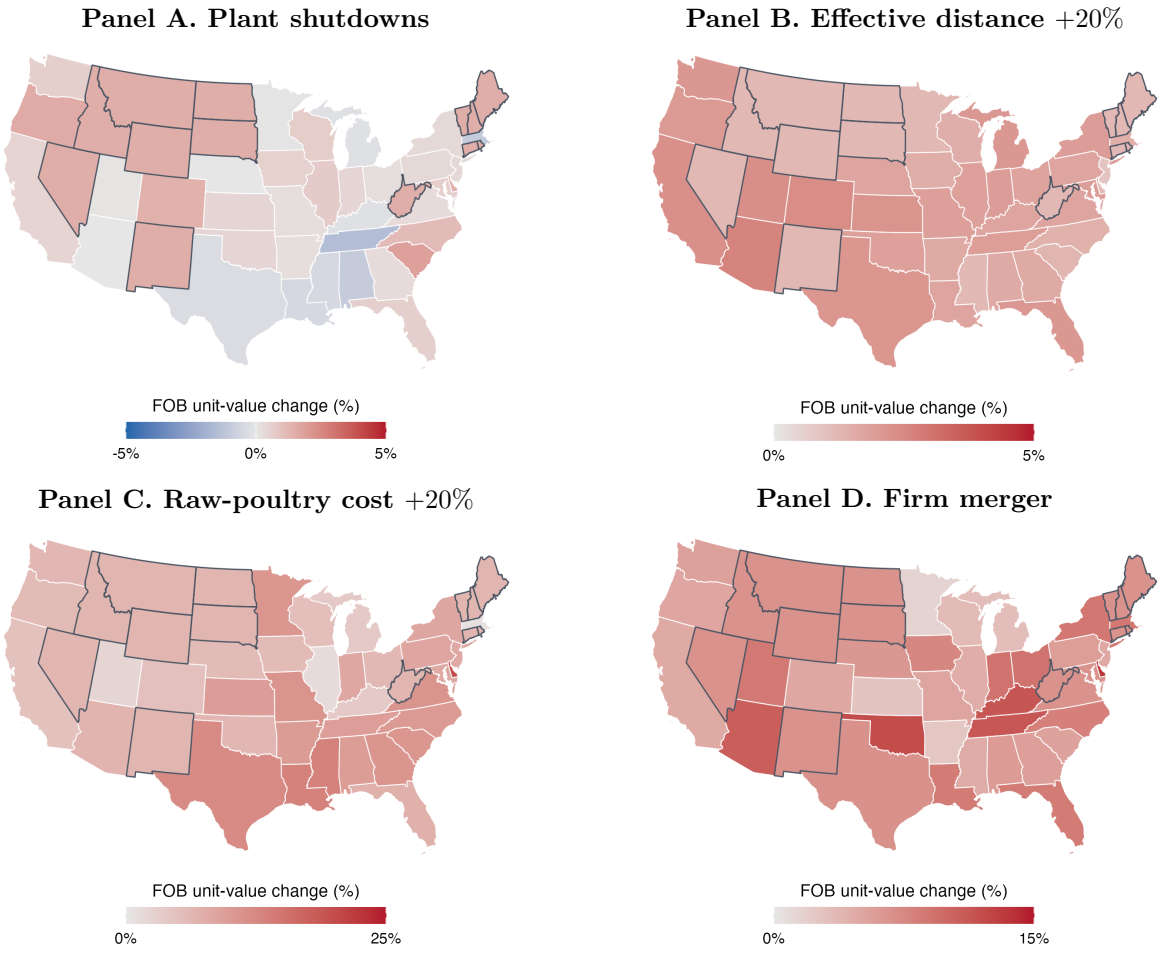


Figure A2: State-Level Price Percentage Changes Relative to the Baseline

Notes: The price is the average equilibrium price in the state row. Panels A–D show percentage changes relative to the model-implied baseline under plant shutdowns in ten confidential counties with a history of flooding, a 20 percent increase in effective shipment distance, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Each panel uses its own color scale with limits rounded outward to multiples of 5 percent. Panels in which all states move in the same direction use a single-hue sequential scale from light gray (no change) to dark red (an increase in the price, cost, or elasticity magnitude, or a decrease in inside sales or surplus) or dark blue (the opposite movement); panels with changes of both signs use a diverging scale centered at light gray. The two inside-sales panels for plant shutdowns and the distance shock share one scale. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

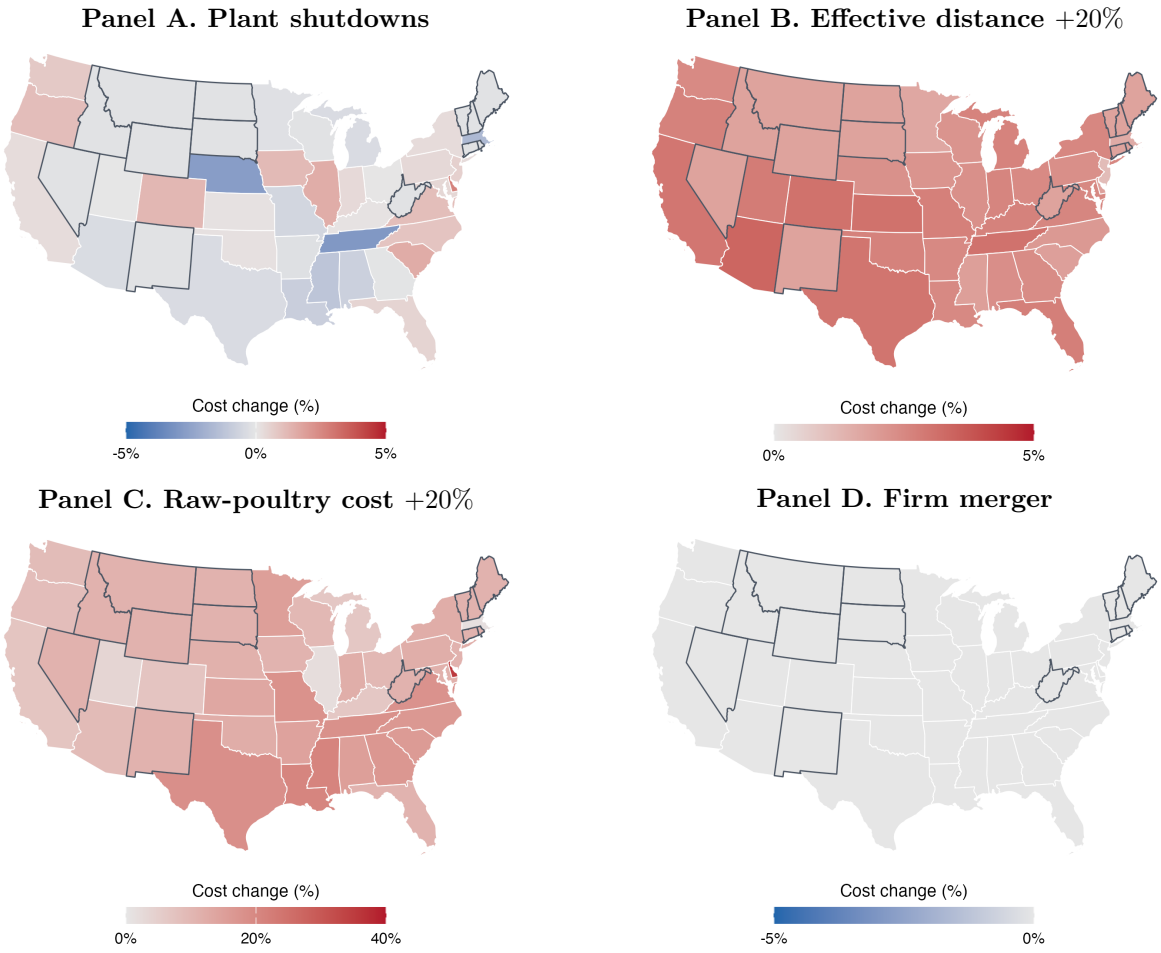


Figure A3: State-Level Cost Percentage Changes Relative to the Baseline

Notes: Cost is the recovered average marginal-cost measure in the state row. Panels A–D show percentage changes relative to the model-implied baseline under plant shutdowns in ten confidential counties with a history of flooding, a 20 percent increase in effective shipment distance, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Each panel uses its own color scale with limits rounded outward to multiples of 5 percent. Panels in which all states move in the same direction use a single-hue sequential scale from light gray (no change) to dark red (an increase in the price, cost, or elasticity magnitude, or a decrease in inside sales or surplus) or dark blue (the opposite movement); panels with changes of both signs use a diverging scale centered at light gray. The two inside-sales panels for plant shutdowns and the distance shock share one scale. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

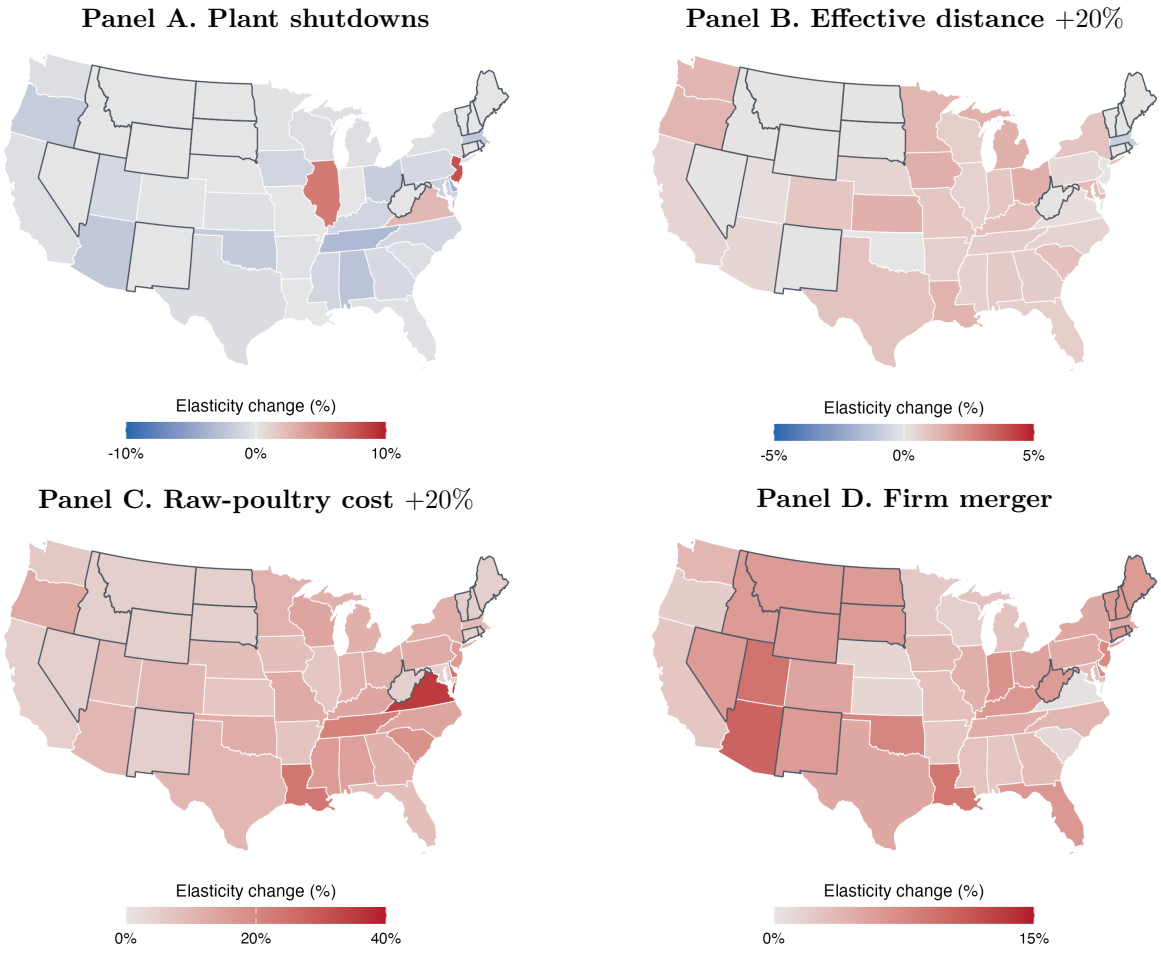


Figure A4: State-Level Own-Price Elasticity Percentage Changes Relative to the Baseline
Notes: Elasticity is the average own-price elasticity in the state row. Panels A–D show percentage changes relative to the model-implied baseline under plant shutdowns in ten confidential counties with a history of flooding, a 20 percent increase in effective shipment distance, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Each panel uses its own color scale with limits rounded outward to multiples of 5 percent. Panels in which all states move in the same direction use a single-hue sequential scale from light gray (no change) to dark red (an increase in the price, cost, or elasticity magnitude, or a decrease in inside sales or surplus) or dark blue (the opposite movement); panels with changes of both signs use a diverging scale centered at light gray. The two inside-sales panels for plant shutdowns and the distance shock share one scale. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

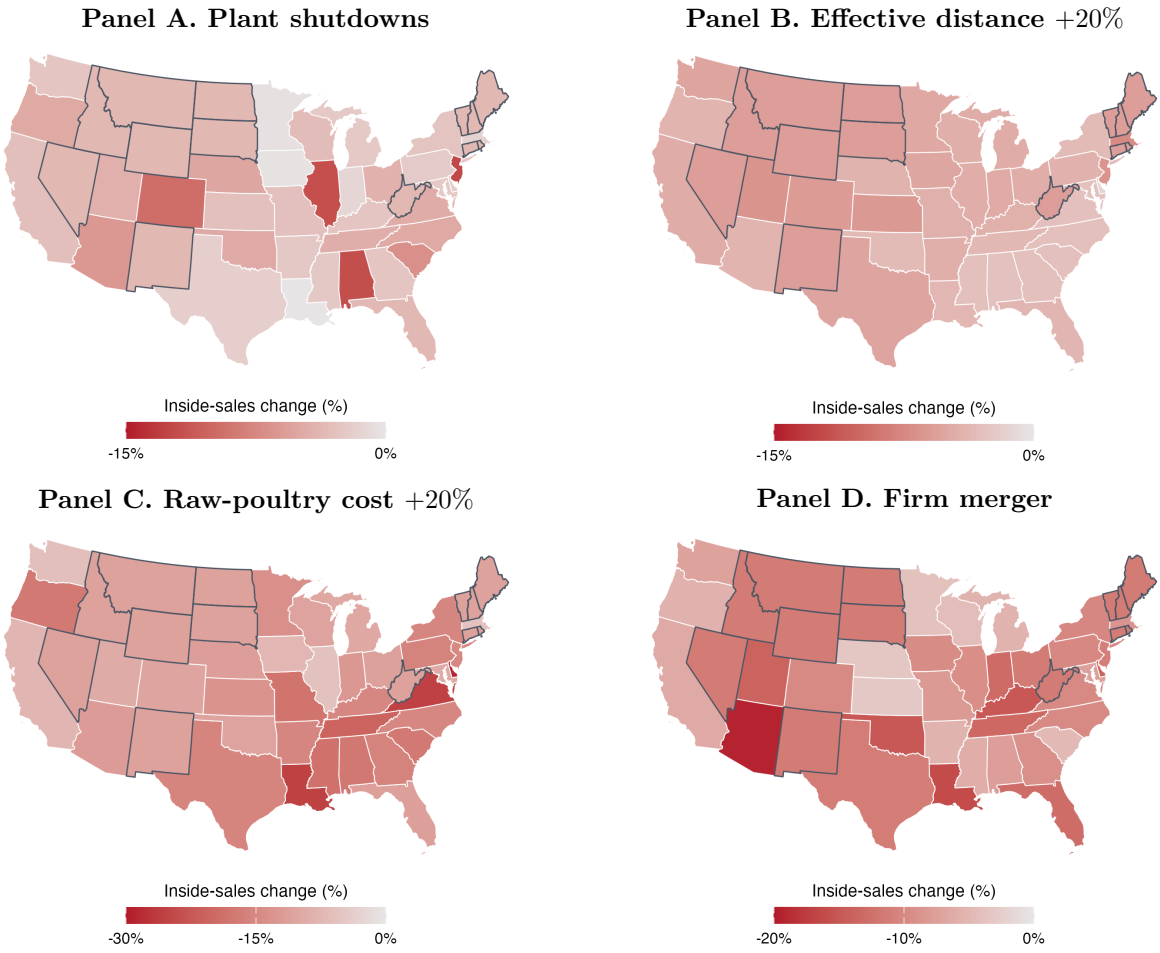


Figure A5: State-Level Inside-Sales Percentage Changes Relative to the Baseline

Notes: Inside sales aggregate the equilibrium inside quantity measure in the state row. Panels A–D show percentage changes relative to the model-implied baseline under plant shutdowns in ten confidential counties with a history of flooding, a 20 percent increase in effective shipment distance, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Each panel uses its own color scale with limits rounded outward to multiples of 5 percent. Panels in which all states move in the same direction use a single-hue sequential scale from light gray (no change) to dark red (an increase in the price, cost, or elasticity magnitude, or a decrease in inside sales or surplus) or dark blue (the opposite movement); panels with changes of both signs use a diverging scale centered at light gray. The two inside-sales panels for plant shutdowns and the distance shock share one scale. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

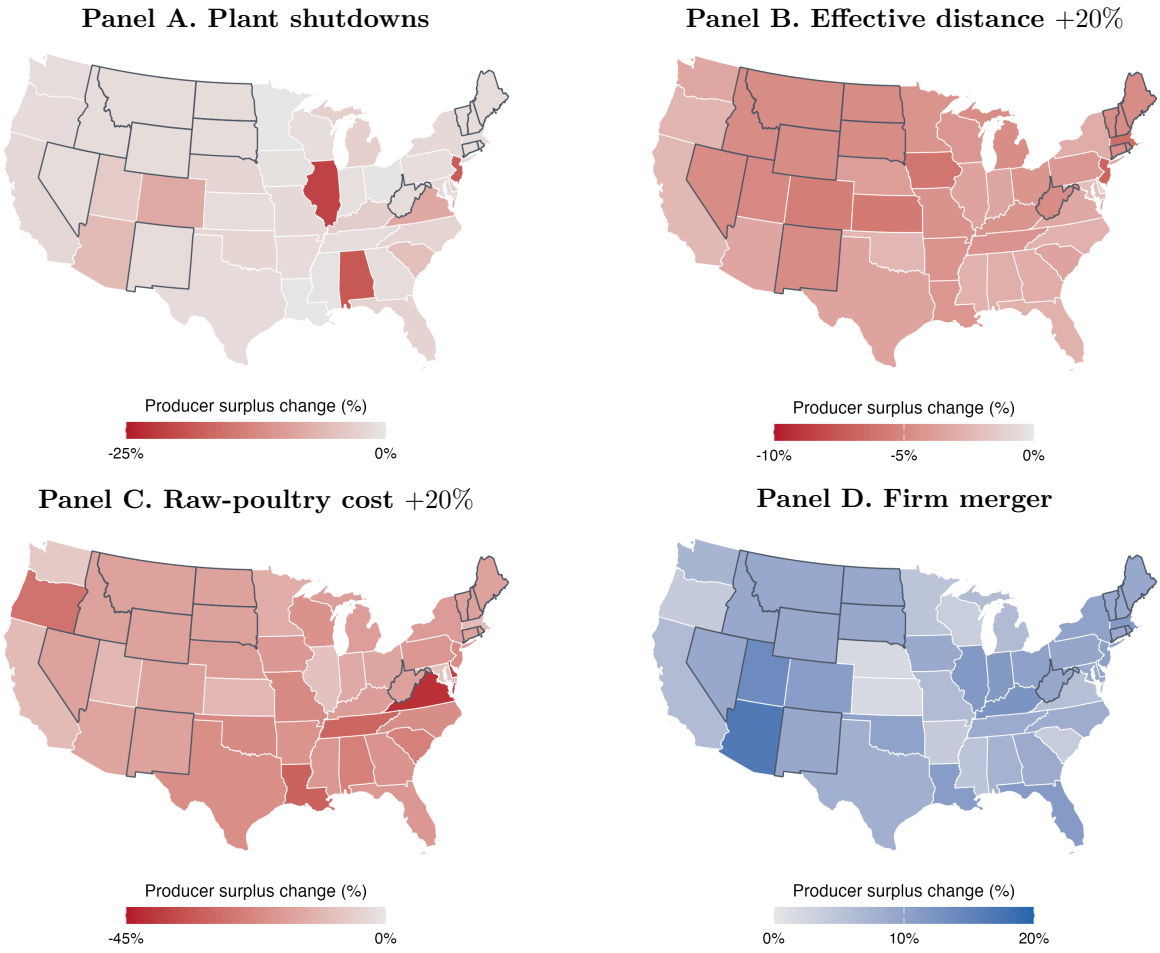


Figure A6: State-Level Producer-Surplus Percentage Changes Relative to the Baseline

Notes: Producer surplus is the model-implied producer surplus in the state row. Panels A–D show percentage changes relative to the model-implied baseline under plant shutdowns in ten confidential counties with a history of flooding, a 20 percent increase in effective shipment distance, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Each panel uses its own color scale with limits rounded outward to multiples of 5 percent. Panels in which all states move in the same direction use a single-hue sequential scale from light gray (no change) to dark red (an increase in the price, cost, or elasticity magnitude, or a decrease in inside sales or surplus) or dark blue (the opposite movement); panels with changes of both signs use a diverging scale centered at light gray. The two inside-sales panels for plant shutdowns and the distance shock share one scale. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

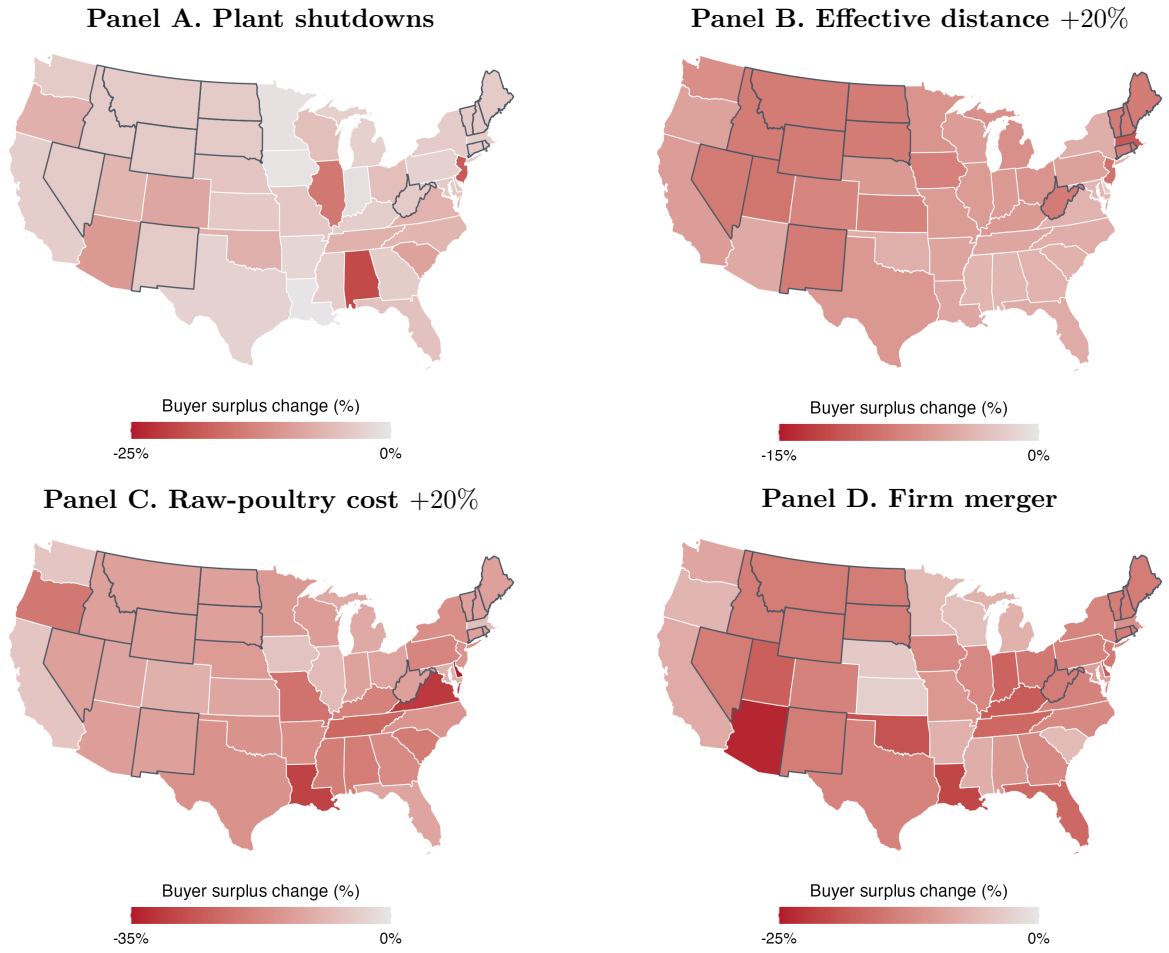


Figure A7: State-Level Downstream-Buyer-Surplus Percentage Changes Relative to the Baseline
Notes: Downstream-buyer surplus is the BLP log-sum surplus measure in the state row. Panels A–D show percentage changes relative to the model-implied baseline under plant shutdowns in ten confidential counties with a history of flooding, a 20 percent increase in effective shipment distance, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Each panel uses its own color scale with limits rounded outward to multiples of 5 percent. Panels in which all states move in the same direction use a single-hue sequential scale from light gray (no change) to dark red (an increase in the price, cost, or elasticity magnitude, or a decrease in inside sales or surplus) or dark blue (the opposite movement); panels with changes of both signs use a diverging scale centered at light gray. The two inside-sales panels for plant shutdowns and the distance shock share one scale. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY26-0215.

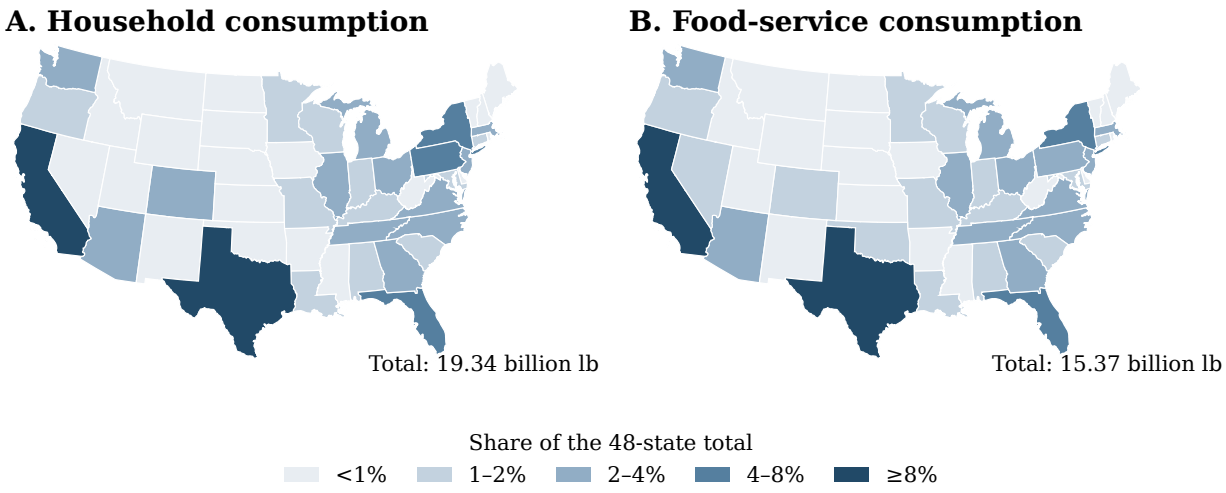


Figure A8: Household and Food-Service Consumption Geography

Notes: State shares of the 2017 retail-weight-equivalent consumption allocations underlying Figure 2. Both panels use the same share bins and normalize within their own consumption category. Household and food-service totals are 19.34 and 15.37 billion pounds. These final-demand locations complement the PR and PW procurement geographies in Figure 3.

B Data Construction Details

This appendix reports implementation details for shipment-channel labels, network statistics, geographic distributions, destination-set coverage, market size, and the LBD sample of procurement decision-makers. These objects support the variables defined in Section 3, but the structural model uses the product-market variables and representative buyer records described in the main text.

B.1 Shipment-Channel Construction

Candidate recipients are classified using establishment industry and, for warehouses, owner-firm industry. Retail stores have establishment NAICS 445 or 452, while food-wholesale centers have NAICS 4244. Warehouses have establishment NAICS 4931. A warehouse belongs to the retail group when its owner firm is classified in 445 or 454 and to the wholesale group when its owner firm is classified in 4244. A destination ZIP containing only retail stores and/or retail-owned warehouses supplies a PR label. A ZIP containing only food wholesalers and/or wholesale-owned warehouses supplies a PW label. Mixed combinations enter the prediction sample. This use of firm affiliation at the destination identifies the warehouse's channel, while the origin establishment's LBD firm identifier supplies the ownership link used in pricing.

The random forest uses shipment size, value, distance, commodity, transport mode, timing, and supplying-plant characteristics and firm affiliation. Destination demographic variables are not predictors. The labeled records are initially partitioned 90–10 for training and validation, with ten-fold cross-validation within the training portion. After model selection, the final classifier is refit on all labels. The full-sample fitted confusion matrix and the approximately 96 percent accuracy of the ten-percent validation sample are separate evaluations.

B.2 Channel Network Statistics

Let e index shipping establishments, z destination ZIPs, $c \in \{PR, PW, From\ W\}$ channels, and y survey years. Denote shipment pounds from e to z in channel-year (c, y) by w_{ezcy} and total outbound pounds by $W_{ecy} = \sum_z w_{ezcy}$. The establishment's destination concentration and the shipment-weighted channel mean are

$$HHI_{ecy} = \sum_z \left(\frac{w_{ezcy}}{W_{ecy}} \right)^2, \tag{27}$$

$$\overline{HHI}_{cy} = \frac{\sum_e W_{ecy} HHI_{ecy}}{\sum_e W_{ecy}}. \tag{28}$$

An establishment serving a single destination has HHI one. More even distribution across destinations lowers the index. This is outbound concentration across destination ZIPs, not concentration of firm ownership among suppliers to a county.

Network density is measured at the county level. Let L_{cy} be the number of distinct positive establishment–county edges, N^E_{cy} the number of source establishments, and N^C_{cy} the number of receiving counties. The bipartite network density is

$$\text{Density}_{cy} = \frac{L_{cy}}{N^E_{cy} N^C_{cy}}. \tag{29}$$

Table 1 averages annual mean prices, shipment weights, distances, and destination HHIs with equal weights over 2007, 2012, and 2017. It sums the annual counts of establishments, destination ZIPs, records, and edges. These sums count appearances in survey years, so a location observed in more than one year can appear repeatedly. The reported density is $\frac{1}{3}\sum_y \text{Density}_{cy}$, not the ratio formed from the summed counts. The three annual networks retain their own source and destination sets.

B.3 State Distributions for the Geographic Comparison

Figures 2 and 3 use a frozen 48-state compilation with factory-production, PR, PW, household, and food-service quantities. Factory production is measured in 2017 ready-to-cook pounds. Its available state entries sum to 46,011,757,000 pounds; the 22 unavailable state entries are set to zero for the displayed state-share comparison. Household and food-service quantities are expressed in 2017 retail-weight equivalents and sum to 19,339,652,264 and 15,369,232,130 pounds, respectively.

For procurement, let Q^{pool}_{sc} be the pooled 2007, 2012, and 2017 quantity for state s and channel $c \in \{PR, PW\}$. The displayed 2017-scaled allocation applies a common national normalization:

$$Q^{\text{geo}}_{sc} = 39.11 \times 10^9 \, \frac{Q^{\text{pool}}_{sc}}{\sum_r \sum_{h \in \{PR,PW\}} Q^{\text{pool}}_{rh}}. \tag{30}$$

The common factor preserves pooled state and channel shares. The PR and PW totals are 18.03 and 21.08 billion pounds after rounding. These geographic allocations are distinct from the annualized plant-product quantities used in estimation. The figures compare state distributions without using the capacity-and-distance links previously selected for presentation maps.

Each distribution is normalized to sum to one across the same 48 states before computing Pearson correlation. The resulting correlations are 0.6874 for production–procurement, 0.1627 for production–consumption, and 0.7130 for procurement–consumption. A common rescaling

of a distribution's total leaves these correlations unchanged. The comparison describes state-share association and supplies no ordering of transport efficiency.

The public processing-location overlay retains plants with FSIS slaughter-volume category 5, labeled at least 10 million birds per 360 days. It contains 164 locations, including 77 plants of the four largest firms and 87 plants of other firms. This display threshold affects the public geographic overlay only. The observed shipment network and structural supplier choice sets retain their own CFS sample definitions.

B.4 Destination-Set Coverage Diagnostic

The structural choice sets use positive processor-origin flows observed in the CFS. I assess how completely the sampled records recover each plant's set of destination counties with a finite-category occupancy model. For plant–survey-year observation e, let n_e denote the number of sampled shipment records used in the diagnostic, let k_e denote the number of distinct destination counties observed among those records, and let $M_e \geq k_e$ denote the number of potential destination counties. The calculation treats the sampled destination labels as independent draws and, as a benchmark, assigns each of the M_e counties probability $1/M_e$. Under this model,

$$\Pr(K_e = k_e \mid M_e, n_e) = \frac{(M_e)_{k_e}\, S(n_e, k_e)}{M_e^{n_e}}, \tag{31}$$

where $(M_e)_{k_e} = M_e(M_e - 1)\cdots(M_e - k_e + 1)$ is the falling factorial and $S(n_e, k_e)$ is a Stirling number of the second kind. Because $S(n_e, k_e)$ does not depend on M_e, the integer maximum-likelihood estimator is

$$\widehat{M}_e = \arg \max_{M \in \{k_e, k_e+1, \ldots\}} \frac{(M)_{k_e}}{M^{n_e}}. \tag{32}$$

I measure count coverage as $C_e = k_e/\widehat{M}_e$. Across plant–survey-year observations with a finite estimate, the disclosure-rounded mean of C_e is about 0.95. This result indicates that, at the county level, sampled shipment records recover nearly all of the effective plant-level destination support implied by the benchmark model.

The equal-probability assumption is useful for making M_e identifiable from (n_e, k_e), but it gives the diagnostic a specific interpretation. If true destination probabilities are unequal, rarely served counties are less likely to appear, and the equal-probability estimate can be smaller than the nominal number of counties with positive probability. At the same time, the observed counties are disproportionately likely to include high-volume destinations. I

therefore interpret $k_e/\widehat{M}_e$ as evidence on coverage of the plant's effective county support, not as proof that every low-probability feasible link is observed. The diagnostic supports defining county-level availability from positive flows while preserving the residual limitation stated in Subsection 3.7.

B.5 Market-Size Construction

The final market-size measure in Subsection 3.5 combines CFS inside shipments, population-based poultry availability, and downstream retail and wholesale activity from ARTS/AWTS. The ARTS/AWTS variables are scale proxies for potential downstream poultry demand; they do not replace the CFS quantities used to form product-level inside shares. I first construct a population-based scale,

$$M_m^{\text{pop}} = \text{pop}_m \times \overline{\text{consumption}}_{t(m)},$$

where $\overline{\text{consumption}}_{t(m)}$ is national annual per-capita poultry availability in the CFS survey year of market m, measured in pounds and constructed from public chicken and turkey availability series (National Chicken Council, n.d.).

Because market m is a destination-county-year cell, let $c(m)$ denote its destination county and let $t(m)$ denote its year. I use county c and year y to describe the auxiliary ARTS/AWTS estimates before mapping them back to market m. For each ARTS/AWTS establishment record i, let w_i denote the survey weight WGT, and let z_i be one of total sales CSAL, total inventories CTINV, or operating expenses COPEX. Records are assigned to counties using reported FIPS codes or a ZIP-to-county crosswalk and are retained for the meat and poultry retail/wholesale NAICS cells used in the downstream calibration. The direct county-year survey estimate for variable z is

$$\widehat{Z}_{cy}^{\text{dir}}(z) = \sum_{i \in \mathcal{S}_{cy}} w_i z_i, \qquad z \in \{\text{CSAL}, \text{CTINV}, \text{COPEX}\}.$$

Annual ARTS/AWTS samples are sparse at the county level, so I stabilize the direct estimates using a five-year window around the year of market m, $\mathcal{W}(t(m)) = \{t(m)-2, t(m)-1, t(m), t(m)+1, t(m)+2\}$. The market year receives weight 3, adjacent years receive weight 2, and the two outer years receive weight 1. Let a_y denote this time weight and let $r_{c(m)y}$ be a reliability weight based on the number of usable survey observations in county $c(m)$ and year y. The pooled direct estimate assigned to market m is

$$\bar{Z}_m^{\text{dir}}(z) = \frac{\sum_{y \in \mathcal{W}(t(m))} a_y r_{c(m)y} \widehat{Z}_{c(m)y}^{\text{dir}}(z)}{\sum_{y \in \mathcal{W}(t(m))} a_y r_{c(m)y}}.$$

For counties with little or no direct survey support, I use the LBD county-year universe to construct small-area predictions. The auxiliary LBD variables are total employment, total payroll, and the number of downstream establishments in the same county-year. On county-years with usable ARTS/AWTS direct estimates, I fit log per-worker outcomes of the form

$$\log\left(\frac{\widehat{Z}^{\mathrm{dir}}_{cy}(z)+1}{\mathrm{EMP}^{\mathrm{LBD}}_{cy}+1}\right) = X'_{cy}\beta_z + \mu_c + \eta_y + \varepsilon_{cy},$$

where X_{cy} contains LBD employment, payroll-per-worker, and establishment-count controls, and μ_c and η_y are county and year effects. The fitted values generate model-based county-year predictions that I map back to each market as $\bar{Z}^{\mathrm{model}}_m(z) = \bar{Z}^{\mathrm{model}}_{c(m),t(m)}(z)$. I combine direct and model-based information with an empirical shrinkage weight,

$$\bar{Z}_m(z) = \lambda_m \bar{Z}^{\mathrm{dir}}_m(z) + (1-\lambda_m)\bar{Z}^{\mathrm{model}}_m(z), \qquad \lambda_m = \frac{n^{\mathrm{win}}_m}{n^{\mathrm{win}}_m + 10},$$

where n^{win}_m is the number of usable survey observations in the five-year window. Finally, I benchmark the county estimates so that their state-year, or national-year when state support is insufficient, totals match the corresponding pooled ARTS/AWTS survey totals. For state $s(m)$,

$$Z^{\mathrm{bench}}_m(z) = \bar{Z}_m(z)\frac{\sum_{c\in s(m)} \bar{Z}^{\mathrm{dir}}_{c,t(m)}(z)}{\sum_{c\in s(m)} \bar{Z}_{c,t(m)}(z)}.$$

The variables in the market-size regression are then

$$\begin{aligned}
\mathrm{meat_sales}_m &= Z^{\mathrm{bench}}_m(\mathrm{CSAL}),\\
\mathrm{meat_inventories}_m &= Z^{\mathrm{bench}}_m(\mathrm{CTINV}),\\
\mathrm{operating_expenses}_m &= Z^{\mathrm{bench}}_m(\mathrm{COPEX}).
\end{aligned}$$

The adjusted market-size proxy is obtained by regressing observed inside shipments on the population and downstream-scale variables,

$$\begin{aligned}
R_m = \zeta_0 + \zeta_1 M^{\mathrm{pop}}_m + \zeta_2 \mathrm{meat_sales}_m + \zeta_3 \mathrm{meat_inventories}_m \\
+ \zeta_4 \mathrm{operating_expenses}_m + \nu_m,
\end{aligned} \tag{33}$$

and denoting the fitted value by $\widehat{M}^{\mathrm{adj}}_m$. The final baseline market-size proxy is

$$M_m = \frac{\max\{\widehat{M}^{\mathrm{adj}}_m, R_m\}}{0.7}. \tag{34}$$

The max operator ensures that market size is not smaller than observed CFS-expanded inside shipments R_m. The denominator is a baseline normalization that leaves room for an outside option, since the retained CFS scope is not a complete census of poultry procurement. I use this market-size proxy to form product shares and the outside share in Subsection 3.5.

B.6 Sampling Procurement Decision-Makers from the LBD

The demand model integrates over 1,000 representative procurement decision nodes in each county-year market. A node represents the choice of an upstream supplier by a food-service organization, retailer, or wholesaler. Physical receipt and procurement control are different: a food-service company can select a processor while a wholesaler receives and delivers the shipment. The three decision-maker types therefore coexist with the two processor shipment channels.

Let a index eligible LBD establishments within 100 miles of market m. Using harmonized NAICS codes in the LBD extract, I classify them as

$$T_{am} = \begin{cases} W, & \text{if NAICS is } 424410, 424420, 424470, \text{ or } 424440, \\ R, & \text{if NAICS is } 445110, 455211, \text{ or } 445240, \\ F, & \text{if NAICS begins with } 7225. \end{cases} \tag{35}$$

Food service is the omitted buyer type in demand. This industry mapping constructs local procurement decision-makers; the destination-ZIP labeling rules in Appendix B.1 instead classify shipment channels.

National throughput per employee converts employment into a procurement scale. For type g, write the public benchmark as $\tilde{\tau}_g = Q_g/E_g$. Food-service and retail poultry pounds use 2017 chicken and turkey per-capita availability and ERS away-from-home shares of 47 percent for chicken and 30 percent for turkey (U.S. Department of Agriculture, Economic Research Service, 2021; National Chicken Council, n.d.). With a population benchmark of 325 million, the corresponding quantities are

$$Q_F = 325 \text{ million} \, (92.1 \times 0.47 + 16.4 \times 0.30) \approx 15.67 \text{ billion lb},$$
$$Q_R = 325 \text{ million} \, (92.1 \times 0.53 + 16.4 \times 0.70) \approx 19.60 \text{ billion lb}.$$

Dividing by Economic Census employment of 10,717,088 for food service and 4,630,685 for retail gives approximately 1,462 and 4,232 pounds per employee. For wholesale, poultry product-line sales of \$24.759 billion are converted to pounds at \$1.498 per pound, giving 16.53 billion pounds. Dividing by 296,581 employees gives approximately 55,728 pounds

per employee (U.S. Census Bureau, 2024c). Relative to food service, these benchmarks are rounded to

$$\tau_F = 1, \qquad \tau_R = 2.9, \qquad \tau_W = 38. \tag{36}$$

These throughput benchmarks use the national public-data calibration above; the geographic panels use the separate state allocation described in Appendix B.3.

I then apply the procurement-control benchmark. Food-service-led and wholesaler-led control receive factors 0.6 and 0.4, while retail receives one. Multiplying these factors by the throughput benchmarks gives

$$a_F = 0.6, \qquad a_R = 2.9, \qquad a_W = 15.2. \tag{37}$$

The 60:40 split is a calibration of who chooses the upstream supplier, not an estimated national shipment share or the household–food-service final-use split. Local employment determines how this calibration translates into each market's mix.

Let $\mathcal{A}_m$ contain eligible establishments within the local area and let $E_{gm} = \sum_{a \in \mathcal{A}_m} \text{emp}_{am} \mathbf{1}\{T_{am} = g\}$. The establishment sampling size is

$$A_{am} = \text{emp}_{am} a_{T_{am}}. \tag{38}$$

The probability of drawing establishment a is

$$\Pr(a \mid m) = \frac{A_{am}}{\sum_{b \in \mathcal{A}_m} A_{bm}}, \tag{39}$$

and the implied type probability is

$$\Pr(g \mid m) = \frac{a_g E_{gm}}{0.6E_{Fm} + 2.9E_{Rm} + 15.2E_{Wm}}, \qquad g \in \{F, R, W\}.$$

The probabilities sum to one within a market. Thus 60 percent food-service-led control does not impose a 60 percent food-service buyer share in every county.

I draw 1,000 nodes with replacement using (39). Each node carries its buyer type and the within-type taste draw in the random-coefficients specification. Since the sampling probabilities already incorporate employment, throughput, and procurement control, the numerical integration weights are equal:

$$\psi_{im} = \frac{1}{1,000}, \qquad i = 1, \ldots, 1,000. \tag{40}$$

The simulated share of a plant-product is the average of its conditional choice probabilities

over these nodes. The 700 estimation markets generate 700,000 nodes. The type probabilities shape integration over demand, while the CFS channel shares describe shipment composition; neither is substituted for the other.

B.7 Instrument Set

The demand system treats the procurement price as endogenous. The excluded demand instruments used in Section 4 are built from three groups:

1. Plant cost and scale shifters: materials, supplemental-input, and labor unit costs; production-worker payroll per hour; production employment; plant asset stock; capital expenditure relative to assets; inventory relative to assets; input-intensity measures for packaging, raw poultry, and poultry parts, defined as the corresponding CMF materials-detail costs per pound of CFS-expanded shipment weight; and indicators for own-slaughter and external-slaughter processing plants.
2. Market-structure shifters: counts of rival products, rival frozen products, rival broiler products, rival processed-poultry products, and distinct firms in the destination-county-year market.
3. Local differentiation instruments: Gandhi–Houde local differentiation measures constructed from frozen status, shipment distance, broiler status, turkey status, and processed-poultry status (Gandhi and Houde, 2019).

The product's own log shipment distance is an included demand characteristic and therefore serves as its own instrument. The distance-based local differentiation measures are separate excluded instruments for the endogenous procurement price. Turkey status is an indicator equal to one for plants whose CMF product line is turkey; because turkey plants belong to the turkey/other-poultry reference group in the demand and supply specifications, the indicator enters only the construction of the differentiation instruments. The supply-side moments fit recovered marginal costs to the included exogenous cost shifters, product and plant-type indicators, shipment characteristics, year indicators, destination population, and rival-count variables in the marginal-cost specification. There is no separate excluded supply instrument set.

The preliminary IV first stages and the BLP demand moments absorb LBD firm, producing-establishment origin-state-by-survey-year, and destination-state-by-survey-year fixed effects. Thus demand-side identification uses the component of each instrument remaining after persistent firm-level differences, shocks common to an origin state in a survey year, and demand conditions common to a destination state in a survey year are removed. The supply-side specification does not impose these demand fixed effects. This conditioning strengthens

the interpretation of cost and scale shifters relative to specifications based only on cross-firm or common regional-year variation, but it does not rule out correlation with time-varying plant-product quality, product composition, county-specific demand shocks, product-specific destination selection, or buyer–supplier matching.

C Variable Definitions

Table A9 defines the variables used in demand, marginal cost, and the preliminary regressions. Their construction is described in Section 3 and Appendix B. The main distinctions are among shipment-based product-market characteristics, procurement decision-makers, plant costs, and market-level demand composition.

A product is a processing establishment crossed with fresh or chilled status (SCTG 05121) or frozen status (SCTG 05122). A market is a destination county in a survey year. This is the establishment–commodity definition with the two retained temperature codes. CMF product-line information separately assigns the plant's commodity group, which determines the broiler, processed-poultry, or turkey/other-poultry nest. Temperature status and commodity specialization consequently enter as distinct characteristics.

Product-market prices, quantities, shipment distance, private-carriage shares, and channel shares are constructed from CFS records using the within-establishment annualization weight in Subsection 3.1. This expansion retains the identity of the observed shipping establishment. The product-market wholesale share ToW_{jm} is PW shipment weight divided by total processor-origin shipment weight; the retail share is $ToR_{jm} = 1 - ToW_{jm}$. The current demand specification uses retail share in levels and interacted with log distance. The marginal-cost specification retains wholesale share and its distance interaction.

Buyer types describe the procurement decision-maker. The indicators type_R and type_W, denoted D^R_{im} and D^W_{im} in the model, identify retail and wholesale decisions; food service is the reference type. Food-service-led procurement can be delivered through a wholesaler, so buyer type and physical shipment channel are separate objects. Let E_{gm} be LBD employment for type $g \in \{F, R, W\}$ within 100 miles of market m. National throughput factors $1 : 2.9 : 38$ are adjusted by procurement-control factors $0.6 : 1 : 0.4$, giving $a_F : a_R : a_W = 0.6 : 2.9 : 15.2$. The type probabilities are

$$\Pr(g \mid m) = \frac{a_g E_{gm}}{0.6E_{Fm} + 2.9E_{Rm} + 15.2E_{Wm}}. \tag{41}$$

The 60/40 benchmark calibrates food-service and wholesale activity weights; the local employment composition determines the resulting market-specific probabilities. The model

draws 1,000 integration nodes from this distribution and averages their conditional choice probabilities. Employment affects the sampling distribution rather than entering as an additional buyer-level preference parameter.

Baseline county PW share measures the wholesale-channel share of processor-origin procurement in the county. It proxies business-demand orientation in the baseline and counterfactual comparisons in Table 4. It is distinct from the product-market channel characteristic and from the probabilities over sourcing decision-makers. A ten-percentage-point comparison refers to a change such as 20 to 30 percent in this baseline share.

The demand specification and its IV first stages absorb LBD firm, origin-state-by-survey-year, and destination-state-by-survey-year fixed effects. Origin state locates the producing establishment; destination state contains the county defining the procurement market. The marginal-cost equation instead uses the cost regressors listed in Table A5. Its moments pair those included regressors with the recovered cost residual.

Plant unit-cost variables mc, sc, and wc are in constant 2017 dollars per pound and represent materials, supplemental inputs, and labor. The data mnemonic mc denotes materials cost; structural marginal cost is mc_{jm}. Rival-count variables summarize the product and firm composition of a destination's supplier set and enter as demand instruments or included cost controls according to the specification. Standard errors for the structural and preliminary estimates are clustered by destination county.

D State-Level Counterfactual Results

Figure A1 first displays the model-implied baseline state-level distributions of the procurement price and cost (both in \$/lb), own-price elasticity, inside sales and market size (both in billion lb), and producer, downstream-buyer, and total surplus normalized by inside sales (in \$/lb). Table A6 reports the state-level rows underlying Figure 5 and the aggregate counterfactual summary in Table 5. As in Section 6, state-level quantity and surplus entries are annual averages of model-implied outcomes in the three CFS survey years (2007, 2012, and 2017); prices, costs, and elasticities retain their quantity-weighted averages. The table reports destination states by name after converting the original destination-state FIPS codes; Rest of America corresponds to the original FIPS 99 aggregate. The table is organized by scenario so that each block can be compared directly with the model-implied baseline. Within a state, the relevant welfare comparison is the change in the TS (total-surplus) column relative to the baseline row for the same state; the main text summarizes these state-level changes in percentage and absolute terms.

In Table A6, Price denotes the procurement price and is reported in \$/lb; cost is also

reported in $/lb; Mkt. and Inside denote market size and inside sales in billion lb per year; and PS, BS, and TS denote producer surplus, downstream-buyer surplus, and total surplus in billions of dollars per year. Within each state row, price, cost, and elasticity are averages across product-market cells weighted by CFS-expanded quantity. The plant-shutdown scenario removes products from ten confidential counties with a history of flooding; the distance scenario increases effective shipment distance by 20 percent; the cost scenario raises raw-poultry cost by 20 percent; and the firm-merger scenario combines the third-largest and fourth-largest firms, ranked by total CFS-expanded shipment value in the estimation sample.

D.1 Public Examples for the Plant-Shutdown Scenario

Table A8 reports public examples of southeastern counties with poultry-processing capacity or upstream poultry production that experienced weather-related disruptions, including floods, storms, and tornadoes. The table is not the confidential county list used to run the plant-shutdown simulation. The examples clarify the type of county used to motivate that choice-set shock; the counterfactual itself remains a stylized simulation rather than an event-study estimate.

E Validation and Sensitivity

E.1 Channel Classification

The PR–PW classifier converts shipment destinations into procurement channels before structural estimation. As described in Subsection 3.2, unambiguous destination compositions supply the labeled records. Model selection uses a 90–10 training-validation partition, with ten-fold cross-validation within the training portion. Validation accuracy is about 96 percent. The final classifier is then refitted on the full labeled sample to classify mixed-destination shipments. Table A1 reports fitted counts from that refit; both channels are recovered despite their unequal frequencies in the labeled records. Shipment and supplying-establishment characteristics are predictors, while destination demographics are excluded. The demographic comparison in Subsection 3.2 provides an additional check on the populations to which these labels are applied.

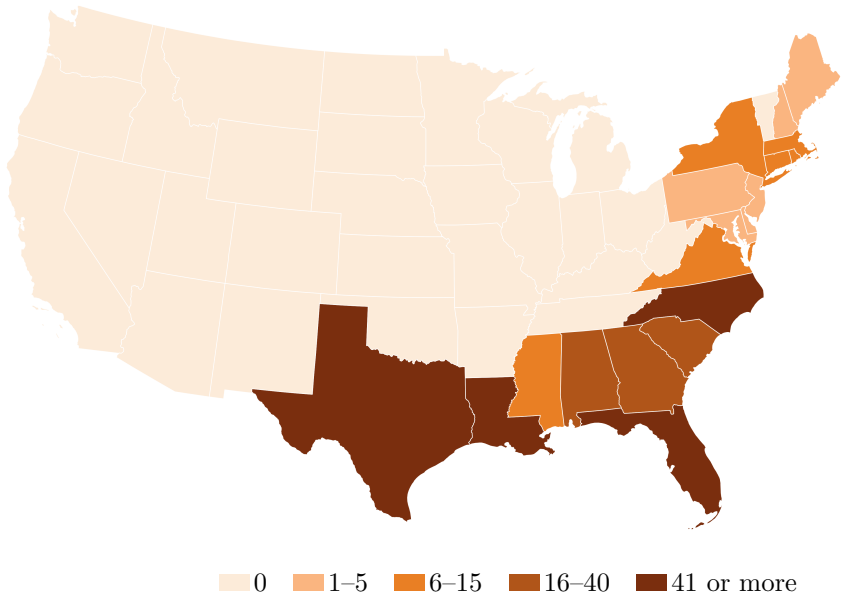


Figure A9: Hurricane Landfalls by State, 1851–2024

Notes: Number of hurricanes (Saffir–Simpson categories 1–5) making a direct hit on each state, 1851–2024, from the NOAA AOML/HRD Hurricane FAQ table of direct hits by state (NOAA AOML/HRD, 2025). A hurricane counts in every state it strikes, so counts cannot be summed across states. Interior states absent from NOAA's coastal table are shown as zero. The leading states are Florida (121), Louisiana (64), Texas (64), North Carolina (58), South Carolina (32), Alabama (23), and Georgia (21). Compare the production panel of Figure 2: the states with the most poultry production also take the most landfalls, which motivates the flood-history county set used in the plant-shutdown experiment.

E.2 Demand Identification

The preliminary demand estimates in Table 2 support instrumenting price and allowing substitution within commodity nests. The IV price coefficient is more negative than OLS, and the nested-logit parameter is positive. The demand equations and first stages absorb firm, origin-state-by-year, and destination-state-by-year fixed effects. Conditional on those effects, the IV logit's cluster-robust Montiel Olea–Pflueger effective first-stage F is 77.62; the nested-logit benchmark's Kleibergen–Paap rk Wald F is 58.19. The former is compared with its effective-F critical value and the latter is reported as a descriptive relevance diagnostic. The exclusion restrictions are the economic restrictions stated in Section 4; the first-stage statistics establish instrument relevance. In the joint model, excluded instruments enter the demand moments, while included cost regressors form the supply moments.

E.3 Market Size, Sampling Depth, and Uncertainty

Table A7 varies the outside-option normalization from the baseline 0.7 to 0.6 and 0.8, re-estimates the model, and resolves all four counterfactuals. Scenario rankings and the signs of the surplus effects persist across these normalizations. Table A3 describes the shipment records underlying the 6,500 product-market observations; cells based on one or two sampled records account for 4.5 percent of expanded shipment weight.

Structural standard errors are clustered by destination county, conditional on the constructed channel assignments, market sizes, procurement-control calibration, and integration nodes. Counterfactual equilibrium totals are evaluated at the estimated structural parameters. The confidence intervals in Table 4 describe the county-level relationships between those outcomes and baseline PW share, controlling for income, education, and population. They quantify the uncertainty of the adjusted county comparisons, rather than sampling intervals for the national counterfactual totals. This distinction keeps the baseline and shock comparisons on the same statistical footing.

F Additional Model and Counterfactual Details

F.1 Preliminary Logit Benchmarks

Following Berry (1994), define the log share relative to the outside option as

$$y_{jm} = \log(s_{jm}) - \log(s_{0m}). \tag{42}$$

The OLS and IV benchmarks estimate

$$y_{jm} = x'_{jm}\beta + \beta_p p_{jm} + \phi_{f(j,m)} + \lambda_{o(j),t(m)} + \eta_{r(m),t(m)} + \xi_{jm}, \tag{43}$$

where β_p is the benchmark's price coefficient and x_{jm} contains the included product and shipment characteristics. The three fixed-effect sets are firm, origin state by survey year, and destination state by survey year. The IV specification instruments price with the excluded demand instruments described in Appendix B.7.

The nested-logit benchmark adds the within-nest share:

$$y_{jm} = x'_{jm}\beta + \beta_p p_{jm} + \rho \log(s_{j|g(j),m}) + \phi_{f(j,m)} + \lambda_{o(j),t(m)} + \eta_{r(m),t(m)} + \xi_{jm}. \tag{44}$$

Commodity nests are defined by the plant's product group; frozen status remains an observed characteristic. Both price and the within-nest share are instrumented. Table 2 reports the price, distance, broiler-distance interaction, and nesting estimates. The price coefficient becomes more negative under IV, and the nested-logit estimate of ρ is 0.448. The effective first-stage F statistic is 77.62 for IV logit, and the Kleibergen–Paap statistic is 58.19 for nested logit.

F.2 Counterfactual Equilibrium and Welfare

For scenario c, I hold $(\hat{\theta}, \hat{\gamma})$, baseline market size, and fitted unobservables for products that remain available fixed. The counterfactual equilibrium solves

$$\mathbf{p}^c_m = \mathbf{mc}^c_m(\hat{\gamma}, \hat{\boldsymbol{\omega}}_m) + \Delta^c_m(\mathbf{p}^c_m; \hat{\theta})^{-1} s^c_m(\mathbf{p}^c_m; \hat{\theta}), \tag{45}$$

following differentiated-products counterfactual methods (Nevo, 2000b). The same-firm indicator in Δ^c_m determines which cross-product sales a firm retains when a buyer switches suppliers. Producer surplus is

$$PS^c_m = \sum_{j \in \mathcal{J}^c_m} (p^c_{jm} - mc^c_{jm}) q^c_{jm}, \qquad q^c_{jm} = M_m s^c_{jm}(\mathbf{p}^c_m; \hat{\theta}). \tag{46}$$

The market size M_m and the integration distribution are held at their baseline values. For representative procurement decision i, the inclusive value of nest g is

$$I^c_{igm}(\hat{\theta}) = \sum_{k \in \mathcal{J}^c_{g,m}} \exp\left(\frac{\delta^c_{km} + \mu^c_{ikm}}{1 - \hat{\rho}}\right). \tag{47}$$

Buyer and total surplus are

$$BS_m^c = M_m \sum_i \psi_{im} \frac{\log\left(1 + \sum_{g \in \mathcal{G}} [I_{igm}^c(\hat{\theta})]^{1-\hat{\rho}}\right)}{\hat{\alpha}_{im}}, \tag{48}$$

$$TS_m^c = PS_m^c + BS_m^c, \tag{49}$$

where $\psi_{im} = 1/1{,}000$. The procurement-control and employment weights enter when the nodes are drawn, and the log-sum is then averaged over those nodes. Buyer surplus measures the value of the modeled upstream sourcing opportunities to the procurement decision-makers.

Price elasticities are

$$\epsilon_{jk,m} = \frac{\partial s_{jm}(\hat{\theta})}{\partial p_{km}} \frac{p_{km}}{s_{jm}}. \tag{50}$$

The diagonal entries measure own-price responses. Off-diagonal entries describe substitution toward other plant-products and, together with their firm affiliations, determine the pricing incentive of a multiproduct processor.

F.3 Annual Aggregation and County Comparisons

Each market m is a county-year observation, so its model-implied quantity and surplus are annual objects. For $Y_m^c \in \{\sum_j q_{jm}^c, PS_m^c, BS_m^c, TS_m^c\}$, the reported national annual average is

$$\overline{Y}^c = \frac{1}{3} \sum_{y \in \{2007,2012,2017\}} \sum_{m:t(m)=y} Y_m^c. \tag{51}$$

State annual averages restrict the inner sum to the state's destination markets. This assigns producer gains to the destinations whose purchases generate them. Prices, marginal costs, and elasticities are averages across the released state rows; dollar and quantity totals are divided by three to place all three survey-year outcomes on an annual-average basis. Percentage changes compare a scenario with the same aggregation of the common baseline. Table A7 reports sensitivity to the market-size normalization.

The county comparisons in Table 4 use a ten-percentage-point difference in baseline PW share as the common measure of business-demand orientation, controlling for income, education, and population. Panel A reports relative differences in baseline procurement and total surplus. Panel B compares losses under the common raw-poultry cost increase. Procurement loss is the reduction in annual pounds purchased from the county's model-implied baseline, and buyer-surplus loss is the corresponding reduction in annual dollars. The 6 and 4 percent contrasts describe relative differences in these loss magnitudes associated with a ten-percentage-point higher baseline PW share. They are not percentage-point differences

in procurement or buyer-surplus decline rates. The table and its intervals retain the sign of the underlying outcome response: the -6 percent procurement contrast denotes a loss 6 percent larger, and the -4 percent buyer-surplus contrast denotes a loss 4 percent larger. The price comparison concerns the additional response associated with PW share, separate from the common price increase under the input-cost shock.

F.4 Shock Implementation

The transportation experiment applies

$$d^c_{jm} = 1.2 d_{jm}, \qquad \log d^c_{jm} = \log d_{jm} + \log(1.2). \tag{52}$$

All demand and cost terms involving log distance, including interactions, are evaluated at the transformed value. The experiment changes the effective friction along every route while retaining plant and destination locations. Because the demand and cost distance slopes depend on product, channel, and year characteristics, their contribution is evaluated for each product-market cell.

For the raw-poultry experiment, let c^{raw}_{jm} denote the raw-poultry input-cost component constructed from CMF materials detail in constant 2017 dollars per pound. The transformation is

$$(c^{\text{raw}}_{jm})^c = 1.2 c^{\text{raw}}_{jm}, \qquad mc^c_{jm} = mc^0_{jm} + 0.2 c^{\text{raw}}_{jm}. \tag{53}$$

Thus the additional raw-input expenditure enters marginal cost dollar for dollar. This shock transformation is separate from the cross-sectional materials coefficient in Table A5; the estimated cost coefficients remain fixed. Processors choose new prices and buyers reallocate procurement in the resulting equilibrium.

The flood-shutdown experiment removes plant-products originating in ten counties with a history of flooding. The county set is fixed before the equilibrium calculations, as described in Appendix D.1. Surviving suppliers can expand at constant marginal cost, and buyers can substitute among those suppliers and the outside option.

The merger combines the third- and fourth-largest firms, ranked by CFS-expanded shipment value, under one firm identifier. Plant-products and marginal-cost primitives remain fixed. Cross-products of the two former firms now have a same-firm indicator of one, so the merged firm accounts for purchases retained across both sets of products when setting prices. All four experiments describe short-run adjustment with no endogenous entry or exit and no binding capacity constraints.